You Can't Sit with Us: How Locals and Tourists Compete for Amenities in Paris[*]

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Abstract

Tourism in cities fosters social interactions between people from distant cultures within limited space. But how does the influx of tourists affect locals' satisfaction with amenities? Using data on restaurant reviews, we construct a panel of tourist presence in Paris. Based on two unanticipated drops in tourism – the November 2015 terrorist attack and the COVID-19 pandemic – we show that tourism reduces Parisians' satisfaction with restaurants. We find that social frictions, including xenophobia towards tourists, drive our results. As tourist numbers declined, explicit complaints about tourists in reviews decreased, while other complaints remained unaffected. Locals are least satisfied with dining among tourists from countries with weak social ties to France. Tourists are not affected by the presence of other tourists.

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1 Introduction

"Are there too many tourists in Paris?" – was the title of a conference organized by the city hall of Paris on June 24, 2019. Experts and officials agreed that overtourism in Paris has not yet reached the same scale as in Amsterdam or Barcelona, but they admitted that "rapid and poorly regulated growth" of tourism can be harmful to the city.[1] There were reasons for concern. The number of foreign tourists to France had more than doubled over the previous 15 years. In 2019, France was the most visited country in the world, and Paris was the third most visited city. During that year, 35.4 million tourists stayed in the city's hotels, which is approximately 16 times more than the population of the city.

In the years before the pandemic, concerns over tourism intensified across Europe.[2] Anti-tourist protests took place in Barcelona, San Sebastián, Mallorca, Venice, and other European cities. Anti-tourist graffiti, typically saying "tourist go home", were spreading across cities, including Paris. However, in the summer of 2020, the crowds vanished. The overtourism debate faded into the background as the COVID-19 pandemic and government restrictions brought global travel to a halt, marking what the World Tourism Organization called "the worst year in tourism history.".

The unexpected shock in tourism created an opportunity to explore the question: "What would life be like for Parisians if there were no tourists?" During the summer of 2020, Parisians experienced a notable absence of tourists, while restaurants and other urban amenities remained accessible – they were kept open artificially through heavy

[1]The World Tourism Organization (UNWTO) defines overtourism as "the impact of tourism on a destination, or parts thereof, that excessively influences perceived quality of life of citizens and/or quality of visitor experiences in a negative way" (Carvão et al., 2018).

[2]See this Guardian article about anti-tourism protest in Europe: https://www.theguardian.com/travel/2017/aug/10/anti-tourism-marches-spread-across-europe-venice-barcelona (last retrieved September 19, 2023).

government subsidies. This provides a unique setting to study demand-related factors without an endogenous adjustment of supply.

In this paper, we aim to estimate the effect of tourism on the subjective quality of life of residents and their satisfaction with urban amenities. Our approach is made possible by the availability of highly granular data on restaurant reviews, which we collected from Tripadvisor – the platform that aggregates user-generated content on restaurant and other travel experiences. We treat this data as a "digital footprint" of urban consumption. First, we use this data to construct a tourism measure at the restaurant level. Second, exploiting information from users' profiles, we identify reviews made by locals, and track individual reviewers over time. Third, we use ratings and texts of reviews left by Parisians as an indicator of their satisfaction with urban amenities.

With this detailed data on urban consumption at hand, we examine two instances of a sudden, unexpected drop in tourist arrivals. As mentioned above, the first episode we study is the COVID-19 pandemic and the associated drop in cross-border travel. This resulted in a dramatic decline in the number of international tourists. The second episode we study is the drop in tourist arrivals caused by the November 2015 terrorist attacks in Paris. These tragic events were widely covered by international news media and in the aftermath Paris was temporarily perceived as unsafe by many. As a result, Paris saw a sizable decline in tourists arriving from abroad, but the impact was smaller than that of COVID-19. The fact that some tourists were still present allows us to test whether tourists themselves are affected by other tourists.

We use a difference-in-differences strategy, classifying restaurants into treatment and control groups based on their pre-shock share of tourist customers. The outcome variable is the average rating by residents per restaurant per month. Our model includes restaurant and month-neighborhood fixed effects to estimate the impact of the tourism decline on satisfaction with "tourist" versus "non-tourist" restaurants. This approach mitigates endogeneity concerns, as both the COVID-19 pandemic and the November attacks could have directly influenced residents' satisfaction and behavior. For instance, after lockdowns, residents may have appreciated public spaces more or sought to support restaurants. By comparing initially more and less touristic restaurants within the same neighborhood over time, we isolate the effect of tourism loss from broader citywide shocks. While these events affected Paris in many ways, their general effects do not threaten our identification. However, our approach assumes that no other factors differentially impacted tourist and non-tourist restaurants beyond the absence of tourists, an assumption we discuss further below.

We find consistent results for both natural experiments: a drop in tourism led to an increase in residents' satisfaction with urban amenities. For the shock brought about by the pandemic, the estimated effect is larger than that induced by the terrorist attack. This aligns with the more substantial decline in tourist arrivals during the pandemic. The estimated effects are statistically significant, and the magnitude is meaningful. Focusing on the COVID-19 results, locals' satisfaction with the average restaurant in the most touristic neighborhoods increases by around 7%.

We repeat this analysis at the review level, allowing us to include user fixed effects to control for changes in reviewer composition. Additionally, we interact user fixed effects with a post-period dummy to compare within-user rating differences between more and less touristic restaurants, accounting for potential selection bias. To validate our amenity measure, we match restaurant data with complaints from the crowd-sourced platform DansMaRue, provided by the Paris city hall. This platform allows residents to report public space issues, such as waste, graffiti, or illegal posters. Using this alternative measure of satisfaction, we find that complaints around restaurants increase with tourism. To further validate our findings, we conduct a placebo test by estimating the effect of the shock on ratings among non-Parisians. As expected, tourist presence does not significantly affect tourists' satisfaction with urban amenities. This test is only feasible for the November 2015 attacks, as tourist numbers were too low after the first COVID-19 lockdown.

In the second part of the paper, we examine three mechanisms behind our findings: overcrowding, supply-side changes, and social frictions. Overcrowding occurs when localized tourist inflows exceed urban capacity, disrupting transportation, increasing pollution, and reducing amenity value. Supply-side changes arise as businesses adapt to tourist demand, often leading to more seemingly local cuisine restaurants, fewer diverse options and potentially lower quality, as tourists are mostly one-time customers. Social frictions stem from locals' attitudes toward tourists, potentially generating a distaste for shared spaces.

We perform several exercises to test the mechanism. First, we rely on text-as-data methods, using the text of restaurant reviews to better understand the subjects of complaints. For this purpose, we identify five common motives present in the reviews: complaints about tourists, low food quality, excessive prices, long waiting times, and noise.[3] We find that tourism increases complaints directly related to tourists, whereas it does

[3] We assign reviews to topics using two approaches: dictionary-based methods and word embeddings. In this study, we use the *word2vec* approach developed in Mikolov et al. (2013), and we mostly follow the procedure of Gennaro and Ash (2022), who performed rhetorical analysis of congress speeches in the US.

not affect complaints about other aspects, such as food quality or overcrowding. This points to a general aversion among residents to dine alongside tourists. Our finding that tourists themselves are not affected by the presence of other tourists is also consistent with this.

Next, relying on a proxy of social connectedness between countries derived from Facebook data, we find that restaurants with a clientele that has more distant connections to France see a larger increase in its rating, once tourists are no longer around. This suggests that Parisians are less bothered by tourists from countries with which they have strong social ties and where social frictions are presumably lower. In line with this, the increase in complaints about the presence of tourists mentioned above is stronger for restaurants having consumers from socially distant countries.

We contribute to the literature on social frictions, discrimination, and segregation in cities by providing evidence of social frictions in the tourist-resident dimension. This complements more traditional results focusing on residents of different races. For example, Davis et al. (2019) show that urban consumption is segregated along racial lines in the US, Algan et al. (2016) study the effects of ethnic diversity of residents on social cohesion at the housing block level. In contrast, we focus on tourism, itself being an important source of social interactions in the city. Given that tourism can be viewed as a facet of international trade, we also contribute to the literature that shows that cultural, social, and ethnic proximity can impact trade patterns (Bandyopadhyay et al., 2008; Guiso et al., 2009; Macchiavello and Morjaria, 2015).

Despite the heated public debate surrounding "overtourism", there is limited causal evidence on the topic. We provide novel evidence that urban tourism has a negative effect on residents' satisfaction with amenities. Thus, we complement an existing literature on urban tourism. Tourism has been shown to have a substantial positive economic impact (Faber and Gaubert, 2019). Lanzara and Minerva (2019) demonstrated a positive correlation between an increasing number of tourists and the number of restaurants and bars. Hidalgo et al. (2022) show that the higher numbers of Airbnb arrivals cause an increase in the number of restaurants. On the other hand, Airbnb has been shown to increase rents (Garcia-López et al., 2020). Finally, Besley et al. (2020) demonstrate how terrorism can negatively impact tourism, supporting our decision to examine terrorism-induced shocks.

We also add to the nascent literature on tourism and amenities by underscoring the significance of social frictions. Almagro and Domínguez-Iino (2022) study how amenities and location sorting by residents endogenously adjust to a large increase in tourist demand, focusing on the city of Amsterdam. In contrast to this paper, we focus on how tourism affects satisfaction with existing amenities rather than how amenities adjust to tourism demand, thereby affecting locals. Allen et al. (2020) study the effects of tourism on residents' welfare in Barcelona using data on consumption patterns and find that tourism expenditure crowds out expenditure by locals. We use detailed restaurant-level data to show how sharing the same amenities with tourists can affect locals' reported quality of life and are able to distinguish different mechanisms through which this effect operates.

More generally, our paper builds on the literature emphasizing the importance of amenities. In their seminal paper, Glaeser et al. (2001) explore the role of cities as centres of consumption. They show that high-amenity cities have been growing faster than low-amenity cities, highlighting the importance of amenities for location choices. Generally, on the importance of urban amenities for attracting residents, see also Carlino and Saiz (2019), Lee (2010), and Couture and Handbury (2020). Finally, we contribute to the diverse literature on the COVID-19 pandemic and its interaction with the city (Althoff et al., 2022; Couture et al., 2022; Coven et al., 2023; De Fraja et al., 2020; Gupta et al., 2022; Miyauchi et al., 2021).

The remainder of the paper is organized as follows. In Section 2, we provide a background on the two tourism shocks that we study. Section 3 describes our main data sources. We explain our empirical strategy in Section 4 and present our main results in Section 5. Section 6 provides robustness checks and additional results. In Section 7, we explore the potential mechanisms behind our main result. Section 8 concludes the paper.

2 Two Episodes of Sudden Tourism Decline

In this paper, we examine two exogenous shocks in tourism that occurred during distinct periods, both of which took place when *Tripadvisor* was in widespread use in Paris. Both shocks provide complementary insights into the impact of tourism on residents' satisfaction with amenities.

2.1 COVID-19 in Paris

The first restrictions related to COVID-19 were implemented in early 2020. On March 12, President Emmanuel Macron announced in a televised address that all schools and universities across France would close. The very next day, on March 13, 2020, Prime Minister Édouard Philippe declared the closure of all pubs, restaurants, cinemas, and nightclubs. After three months of strict lockdown measures, cafes, restaurants, and pubs fully reopened in Paris on June 15, following a brief two-week period where only outdoor seating was possible.[4]

While the restaurant sector began its return to normalcy, tourism continued to suffer heavily due to the global pandemic. The Île-de-France region, which encompasses Paris and its suburbs, was particularly hard-hit. Compared to July 2019, overnight stays in its hotels in July 2020 decreased by 70.8%.[5] Subsequent months witnessed a comparable downturn in the hospitality sector. This decline was notably accentuated among tourists not residing in France. In 2020, France experienced a 71.8% reduction in non-resident overnight stays compared to 2019, whereas overnight stays by residents decreased by only 10.5%.

The drop in tourism and COVID-related restrictions led to a large shock to restaurant revenue.[6] In response to this looming crisis, the government implemented safety nets in place that ensured the survival of most affected businesses. Bankruptcies actually declined dramatically during the pandemic. According to the Ministry of Finance, in the food service sector, business closures were down 57% from March 2020 to October 2021 compared to the pre-pandemic period.[7] The composition of restaurants thus stayed remarkably stable over this period of changing demand. This provides us with an ideal setting to study the effects of competing for access to existing amenities.

[4]Some sanitary measures remained in place. Hygiene protocols had to be respected. For indoor dining a minimum distance of one meter between tables was required and only a maximum group size of 10 people was allowed.

[5]See this article by the French national statistical agency INSEE: https://www.insee.fr/fr/statistiques/5369851 (last retrieved on September 15, 2023).

[6]In Figure A.12 we document a sharp drop in activity around retail and recreation locations during the lockdown period, as measured by Google's Mobility Index.

[7]See this report for further details: https://www.tresor.economie.gouv.fr/Articles/2022/01/18/business-failures-in-france-during-the-covid-19-crisis (last retrieved on September 15, 2023).

To further illustrate the nature of the shock, Figure 1 shows the number of reviews written in French and other languages, separately. The beginning and the end of the "first-wave" lockdown imposed by the French government are marked with a blue dotted line. During the lockdown both French and non-French reviews dropped to near zero. Then, starting in June, French reviews revived, but foreign reviews remained on a negligible level. The observational period ends with both French and non-French review numbers going back to zero due to the introduction of a second wave of restrictions. This demonstrates that demand by tourists remained low during the summer after the first lockdown while locals quickly returned to restaurants.

2.2 November 2015 Paris attacks

On November 13, 2015, Paris was hit by a series of terrorist attacks. The gunmen fired at civilians in multiple locations across the city, leaving 130 dead. This gruesome attack immediately captured both national and international media attention. Detailed accounts of survivors were quickly spread across the globe. This indiscriminate attack on civilians made Paris an unsafe place to visit in the eyes of some and led to a substantial decline in tourism. For example, overnight stays in Paris by visitors from foreign countries dropped by 9.8% year-on-year in the fourth quarter of 2015.[8]

Figure 2 shows the decline in non-French reviews after the attacks, with no discernible impact for French reviews. The November attacks thus lead to a sudden decline in tourism. However, tourism did not decline as sharply as during the pandemic. This allows us to see how satisfaction with amenities evolved among those tourists that came

[8]See this article published by the French national statistical agency INSEE: https://www.insee.fr/en/statistiques/2011367 (last retrieved September 19, 2023).

despite the shootings.

3 Data

In this section we first discuss our primary dataset on restaurant reviews collected from the website *Tripadvisor*, describe our measure of tourism and briefly introduce the text analysis we perform on the review texts. Finally, we describe additional datasets from other sources that we use.

3.1 Tripadvisor Data

Tripadvisor is a user-generated social media review site, which publishes user reviews on restaurants, hotels and other attractions. We collected data on all Parisian restaurants that were listed on the site on November 17, 2020.[9] We obtained information on restaurant characteristics, such as the type of cuisine and the address, and individual review data, including the review's date, text, language, user, user location and rating. We geocode restaurants' addresses. We leverage the data on review's language and user location to separate consumption of residents and tourists. As a result we construct a unique and highly detailed panel that reflects the city's restaurant consumption across space and time. French users began adapting the platform in 2007, and their usage peaked in 2017.

The entire set of Parisian reviews that we collected from 2008 to 2020 consisted of around 2 million reviews for approximately 15,000 restaurants, cafes, and bars. We then sampled and preprocessed them for further analysis. We assign each restaurant to one of Paris' 80 administrative neighborhoods (see A.3 for a neighborhood map).

[9] In this analysis, we restrict ourselves to restaurants located in Paris *intra-muros* – the city of Paris that consists of 20 municipal arrondissements and excludes the surrounding Greater Paris area.

While digital platforms are ubiquitous and widely used, their active users might not be representative of the general population. However, we argue that TripAdvisor data, if not representative, captures a sizeable and meaningful segment of the population. While we do not have demographic data on individual users, aggregated data suggests that the gender split is roughly balanced on TripAdvisor with 55% of users estimated to be women (similarweb, 2024).[10] As is generally true for users of digital platforms, TripAdvisor users are younger than the average Parisian. The largest group of users among adults is aged between 25 and 34 with 27.7% (see Table B.3). Nevertheless, a sizeable 13.9% of users are estimated to be aged between 55 and 64, and the average age of users (41.5) is remarkably close to that of Parisians (39).[11]

TripAdvisor is a leading platform for dining experiences. Google Trends shows that TripAdvisor often outperforms its closest competitors, in particular a local competitor called *La Fourchette* (see Figure A.13). Unlike other platforms, TripAdvisor has been primarily focused on reviews. Users are often invested in leaving comprehensive reviews.

3.2 Measuring Tourism

In this paper we use review data to construct a highly granular measure of tourism at the restaurant level. Importantly, it gives us an indicator of where tourists consume in the city rather than where they stay. Our preferred proxy of tourism is constructed as a share of reviews written in languages other than French. In Section C in the Appendix we repeat our analysis using an alternative measure of tourism based on users' home

[10] These numbers are not based on user-reported information, but are constructed based on user behavior combining various third-party data sources such as Google Analytics. They only provide a rough approximation.

[11] The remaining age groups are as follows: 18-24: 11.6%, 35-44: 20.5%, 45-54: 18%, 65+: 8.1%.

locations.

The Figure 4 shows a map of our tourism measure. A lighter color indicates a higher share of non-French reviews. As expected, restaurants with the highest levels of tourism are located in the areas known for Paris' major attractions: the Eiffel tower, Montmartre, Notre-Dame de Paris and the Arc de Triomphe.

To validate our proxy for tourism more formally, we use data from the *Enquêtes de fréquentation des sites culturels* (Attendance surveys of cultural sites) provided by the *Observatoire économique du tourisme parisien* (Observatory of the Parisian tourism economy). This survey details the proportion of all tourists coming to Paris who visit various tourist attractions. We consider tourists visiting from 2015 to 2019 and geocode the 18 attractions that are located intra-muros contained in the survey. Then, we construct a measure for demand by tourists that follows the market access framework widely used in the economic geography literature:

$$\text{Tourist Access}_i = \sum_j \frac{\text{Visitors}_j}{\text{Distance}_{ij}},$$

where i is a restaurant and j is a tourist attraction. We are implicitly assuming a distance elasticity of tourist consumption trips of -1. While we are not aware of a paper estimating this parameter specifically for demand by tourists, Miyauchi et al. (2021) look at the distance elasticity of location choice for consumption trips. They find a value of -1.09 and thus close to -1.

Next, we correlate our tourism proxy with the tourist demand measure. As Figure A.6 shows, we find a strong positive correlation between the two (the R^2 of a linear regression is 0.19). The correlation is robust to controlling for neighborhood fixed effects, meaning that, even after controlling for a relatively fine-grained spatial unit, the remaining variation in our tourism proxy is correlated with tourist access (see Table C.1). Together, this shows that our proxy for tourism correlates strongly with other, external measures of tourism.

To further corroborate our proxy for tourism, we rely on user location information. In particular, we compute the share of users by restaurant who indicate a location in a country other than France. As Figure A.7 shows, the two measures are highly correlated (the R^2 of a linear regression is around 0.77).

For a final validation exercise, we compare the reviewing behavior of tourist users (those writing reviews in a foreign language) and locals. As tourists typically consume a lot in a relatively short amount of time, we expect the time between reviews to be shorter for tourists than for locals. Indeed, we find that the median time between reviews is 1 day for tourists and 24 days for locals.

3.3 Content of Reviews

We perform text analysis of reviews to better understand users' concerns. We distinguish five potential disamenities we want to test for: discussion on the presence of tourists themselves, concerns about low food quality, high price, long waiting time and noisy environment.

To identify these concerns in the text, we employ two established approaches from the literature: a dictionary-based method and word embeddings. In the dictionary-based method, we pre-select a set of words and phrases associated with the relevant topics. For word embeddings, we adhere to the technique proposed by Gennaro and

Ash (2022), which involves calculating cosine distances between the related dictionary centroids and the centroids of text reviews in word2vec embeddings. Each approach has its own advantages and drawbacks. In this section, we detail these methods and explain how we constructed each measure.

3.3.1 Dictionary-Based Approach

The mapping of review texts to topics is achieved through manually constructed dictionaries. The construction procedure unfolds as follows. First, we examined approximately one thousand randomly selected reviews to identify words that unambiguously pertain to the topic. Second, we validated these terms by searching for counter-examples within the corpus to highlight the "false positives" – reviews where these terms appear, but are not genuinely related to the topic. Third, we augmented our dictionary to include common misspellings and partial forms of the chosen terms. Lastly, we assembled a list of 'minus' phrases to ensure that expressions like *"pas cher"* (not expensive) are not mistakenly flagged as *"cher"* (expensive).

In essence, our methodology primarily minimizes *false positives* (the risk of erroneously attributing a text to a topic when it is not pertinent), but it does not actively reduce *false negatives* (the chance of overlooking a text's relevance to a topic). The concise version of our dictionary (excluding misspellings and variations) can be found in Appendix Table D.1. The summary statistics of the topics are included in Appendix Table B.1. It is noteworthy that all topics manifest with relatively comparable frequencies (between 2% and 6%).

3.3.2 Word Embedding

To complement our dictionary-based measure, we adopt the word2vec method (Mikolov et al., 2013). This model represents words as vectors, often referred to as embeddings, that refine the extensive and sparse co-occurrence information throughout the corpus into more concise, lower-dimensional forms. In this generated space, the "semantic" distances between words can be assessed. This technique has been effective in many social science applications. Gennaro and Ash (2022) outline a method for using word2vec in analyzing economic data. We follow their guidelines for preprocessing, but make an exception for parts of speech selection, as we believe our topic might require the inclusion of additional linguistic elements beyond just verbs, subjects, or objects.

This approach offers the advantage of extracting more nuanced information from texts than simply relying on the presence of predefined keywords. Therefore, our choice of such semi-supervised approach allows us to conduct hypothesis-driven analysis without sacrificing generality. This approach enables us to define interpretable axes – such as quality of food, overcrowding, long waiting times, excessive prices, and concerns about tourists – that align directly with our research questions[12]. For a detailed look at this seeding vocabulary, please refer to the Appendix Table D.2.

It is important to note that, as shown in Figure 8, our defined topics – measured using cosine distances in the embedding space – demonstrate meaningful correlations. These topics can be logically grouped according to the underlying mechanisms. For instance, the most correlated variables, such as long waiting times and noise, are both associated

[12]The embeddings-based approach allows us to define axes flexibly. However, a limitation is that the choice of topics is not entirely data-driven. In a topic modeling implementation, which is available upon request but not included in the paper, we show that methods like LDA were less effective in capturing the dimensions of interest, such as complaints about tourists or overcrowding, instead identifying topics unrelated to our hypotheses (e.g., cuisine types or generic quality evaluations).

with the overcrowding mechanism.

3.4 Data from the Application "DansMaRue"

Most of our analysis is based on the TripAdvisor data. To externally validate that the presence of tourists affects locals' satisfaction with amenities, we draw on an additional dataset from the application *DansMaRue* created by the Municipality of Paris. With the help of this application, citizens can register and geolocalise 'anomalies' observed in public space in Paris.[13] Users upload the complaints directly from their smartphones, specifying the location, date and the subject. The aim of the application is to improve the quality of Parisian public space by giving access of user-generated data on 'anomalies' to municipal service. The application was launched in 2012. For our analysis we focus on complaints about commercial activity which is the category most related to restaurant activity.

The high resolution of the data allows us to only consider complaints that are possibly related to a particular restaurant. We assign complaints to a given restaurant within a 100m radius or to a neighborhood, depending on the specification.

3.5 Social Connectedness Index

Below we want to test whether the origin of tourists has an impact on locals' perception of them. To proxy for social (or network) proximity between foreign countries and France we rely on the Social Connectedness Index (SCI) published by Facebook.[14] It is based on the number of Facebook friendships between users located in a pair of countries. More

[13] The set of potential 'anomalies' includes overflowing litter bins, illegal graffiti, abandoned objects, road damage and many others.

[14] The version we use dates from October 2021.

precisely, it is computed as

$$\text{Social Connectedness}_{ij} = \frac{\text{FB Friends}_{ij}}{\text{FB Users}_i \times \text{FB Users}_j},$$

where FB Friends_{ij} are the number of friendships between users residing in countries i and j and FB Users_i the number of users in country i. The central premise of the index is that the 'network' connectedness between regions and countries acts as a measure of familiarity among their residents and the closeness of their social relationships. Social connectedness is influenced by cultural, geographical, political, and historical ties. For further details on the methodology see Bailey et al. (2018). Relying again on the information on users' origin, we compute the average social connectedness between the French population and the non-French customers of a particular restaurant.

3.6 Descriptive Statistics

During our main study period, spanning January 2018 to November 2020, our TripAdvisor dataset includes approximately 11,000 restaurants and 112,000 reviews left by locals. On average, a restaurant has a 19% probability of receiving at least one review in a given month, with the average rating from Parisian customers being around 3.9. There are around 45,000 Parisian users that left at least one review during the study period. Table B.1 provides further descriptive statistics.

As anticipated, reviews are not equally distributed across restaurants and users. Figure A.2 presents histograms of review counts by restaurant and user for the two episodes analyzed. The top-left panel illustrates that some restaurants were reviewed over 1,000 times during the study period, while others were reviewed only once. Similarly, certain users left hundreds of reviews, whereas others contributed just a single review. This skewed distribution is common for user-generated social media data. To ensure the robustness of our findings, we repeat our main analysis, excluding both the most-reviewed restaurants and the most active users.

The average share of non-French reviews, which serves as our primary indicator of tourism, is 0.28. In Figure 3, we categorize restaurants into four groups based on their degree of tourism and calculate the average rating for each group over time. Before the pandemic, the most touristic places – those in the top 25% – had significantly lower average ratings. However, after the onset of the pandemic, this gap diminished, and the groups became largely indistinguishable. This provides descriptive evidence suggesting that in the absence of tourists, Parisians began to appreciate previously touristic locations more.

We present descriptive statistics at the user level in Table B.2. Our analysis employs progressively more stringent specifications, with some relying on user-level fixed effects. As a result, the variation may stem from different samples of users; however, these samples are not drastically different, as shown in Table B.2. In summary, approximately 45,000 users left at least one review during the pre-lockdown period (January 2018 to February 2020), and around 7,000 users did so during the (shorter) post-lockdown summer of 2020 (June to November 2020). About 17,000 users left at least two reviews across the entire period, with 3,000 users leaving reviews both before and after the lockdown.

4 Empirical Strategy

We want to study how the sudden reduction in tourism brought about by the COVID-19 pandemic affected residents' satisfaction with restaurant visits. Using the *TripAdvisor* dataset, we employ a standard difference-in-differences framework at two different levels of aggregation. First, we aggregate our data up to the restaurant level. We thus study how locals' average rating changes within a restaurant across time, and if it does so differentially across more and less touristic venues. Second, more granular, review-level regressions allow us to compare variation within user across time. Hence, we asses whether the same users evaluated initially more touristic restaurants differently when borders were closed. Our identification assumes that the shocks only impact tourist and non-tourist restaurants differently through the absence of tourists. We examine the validity of this assumption below.

Although *TripAdvisor* data provides a rich perspective on urban consumption, its structure poses certain challenges. Ideally, we would observe the same user reviewing the same restaurants multiple times, allowing us to capture time-invariant effects at the restaurant-user level. In practice, most users review a restaurant only once, resulting in a dataset dominated by unique restaurant-customer pairs – an arrangement that can introduce compositional shifts. Nevertheless, we conduct a series of robustness exercises that address this concern and further reinforce the reliability of our findings.

4.1 Restaurant-Level Analysis

At the restaurant level, we use the following specification:

$$\mathrm{Y}_{jt} = \beta \times \mathrm{Post}_t \times \mathrm{Tourism}_j + \gamma_j + \delta_t + \epsilon_{jt}, \tag{1}$$

where Y_{jt} is an outcome of restaurant j in month t. Our explanatory variable here and in the following specifications is $\mathrm{Post}_t \times \mathrm{Tourism}_j$. Post_t is a binary variable indicating whether month t belongs to the post-lockdown (or post-attack) period. $\mathrm{Tourism}_j$ is our treatment exposure variable, indicating the extent to which restaurant j is frequented by tourists prior to the shock. It is measured by the share of non-Parisian reviews prior to a shock. Standard errors are clustered at the neighborhood level.

We include restaurant fixed effects (γ_j) and month fixed effects (δ_t). This baseline specification thus accounts for any cross-sectional variation such as more touristic restaurants being of worse quality. In addition, any aggregate changes in review behavior after the pandemic are accounted for, such as a general fear of contracting the virus leading to generally lower satisfaction with dining out.

In a more stringent variation of this specification we also include neighborhood-time fixed effects. This controls for any unobserved time-varying factors at the neighborhood level, such as an increased share of remote work during COVID-19, that may affect residential neighborhoods differently than the business district. Also, Covid-19 brought a reduction in demand for short-term Airbnb rentals, potentially impacting neighborhoods differently. Our specifications with time-varying neighborhood effects account for this possibility.

Below we will focus on one main outcome. We look at the average rating that restaurant j receives in month t, only looking at reviews by residents. Our hypothesis is that tourism lowers the utility residents derive from amenities. We thus expect $\beta > 0$, locals becoming more satisfied with initially touristic restaurants once tourists are largely absent from the city.

4.2 Review-Level Analysis

The granularity of our data allows us to run additional analysis at the review level. We use the following specification:

$$\mathrm{Y}_{ijt} = \beta \times \mathrm{Post}_t \times \mathrm{Tourism}_j + \gamma_j + \delta_t + \mu_i + \epsilon_{ijt}, \tag{2}$$

where Y_{ijt} is a rating by user i for restaurant j in month t. Our explanatory variable is the same as above. Again, we cluster standard errors at the neighborhood level.

Importantly, in addition to restaurant and month fixed effects (γ_j, δ_t), we also include user fixed effects (μ_i). We thus rely on within-user variation across the pre- and post-lockdown periods, asking if the same user rated touristic places differently, once tourists were no longer around.

While including user fixed effects is already highly restrictive, in a final step we interact user fixed effects with a post-lockdown dummy. This controls for changes from the pre- to the post-lockdown period at the user level, such as health shocks. Intuitively, this specification captures whether the penalty for more tourist places decreased after the lockdown relying only on different ratings for more or less touristic restaurants by a single user in the same period.

Our parameter of interest is β. Our hypothesis is that tourism is detrimental for residents' utility derived from a restaurant visit. Hence, we should observe that post-lockdown, when restaurants were open, but tourists were not present, initially tourist places start receiving higher ratings ($\beta > 0$).

5 Main Results

5.1 Pandemic-Induced Tourism Decline

Columns 1 and 2 of Table 1 show the results of estimating Equation 1 using the average monthly rating by Parisians at the restaurant level as the outcome variable and comparing the periods before and after the first COVID-19 lockdown. The sample period goes from January 2018 to November 2020. The results show that initially more touristic venues receive higher ratings by locals when tourists are no longer around. Importantly, as shown in Column 2, this effect is not driven by neighborhood-level trends as including neighborhood-time fixed effects only marginally changes the coefficient.

To better understand the magnitude of our estimates, consider a restaurant located in the area surrounding the Notre-Dame cathedral, a major tourist attraction. The average share of non-French reviews in this area is 64%. The estimates in Column 2 imply that locals' rating for this restaurant would increase in the absence of tourists by around 0.05 on a scale from 0 to 1, or by 7% relative to the mean. Tourism thus causes a substantial decrease in locals' satisfaction with amenities.

Table 2 shows the results of a user-level estimation (see Equation 2). Columns 1 to 4 in the first panel confirm our results at the user level, i.e. Parisians rate their experience higher in places previously frequented by many reviewers not from Paris. The coefficient is of similar magnitude as at the restaurant level. Importantly, in Columns 2 to 4 we include user fixed effects, exploiting within-user changes in behavior while holding fixed time-invariant individual characteristics, such as preferences for certain types of neighborhoods or restaurant types.

The nature of restaurant reviews does not allow us to control for user-restaurant fixed effects, since the vast majority of users rates a restaurant only once. However, we can allow for the user fixed effect to vary between the pre- and post-lockdown period. If, for example, there is an unobserved factor that causes both a mood shift among users and a systematic change in visiting more (or less) touristic restaurants, this could bias our results. Including a user fixed effect interacted with the post-lockdown dummy accounts for this possibility.

5.2 November 2015 Attacks

As argued above, the pandemic provides us with a sudden decline in tourism demand, while leaving existing amenities mostly intact. Still the pandemic may have affected restaurants in ways that are unobservable to us and correlated with our measures of tourism. For example, restaurants with larger outdoor facilities may have benefited most after the lockdown was lifted, as people continued to be cautious because of the risk to get infected. If the availability of outdoor facilities is correlated with our measure of tourism, we are wrongly attributing the observed changes in ratings and demand to tourism.

To alleviate concerns related to the specific nature of the pandemic, we instead use the terrorist attacks that took place on November 13, 2015 as an exogenous shock to tourism. Columns 3 and 4 in Table 1 display the results of estimating Equation 1 using reviews from November 2014 to November 2016.[15] We find that initially more touristic

[15] We define tourism intensity based on data from 2014. November 2015 is dropped from the analysis and December 2015 onwards is defined as post-attacks.

restaurants received better ratings by Parisians after the November attacks. Compared to the lockdown-related shock, the coefficient is smaller, which is in line with a lower drop in tourism arrivals than during the summer of 2020.

While we conduct restaurant level analysis with the terrorism shock, applying a review-level analysis here is too demanding. We provide the review level specification for the terrorist attack shock in Appendix Table B.6. The results are noisier. While we find a statistically significant effect in the first column, we estimate a positive, but insignificant coefficient when controlling for user fixed effects. As explained previously, the size of the tourism decline caused by the November was smaller, and likely dissipated faster as media attention faded.

Overall, this very different natural experiment lends support to our hypothesis that tourism negatively affects the quality of amenities as perceived by locals. The result does not seem to be driven by factors specific to the pandemic.

Finally, the November attacks allow us to look at the reaction of reviewers that are not from Paris. As shown in Columns 5 and 6 of Table 1, there is no effect on their ratings of touristic places. Externalities caused by tourism specifically affect locals. This suggests that general disamenities such as congestion are unlikely to be at play, but rather the presence of tourists themselves bothers locals.

6 Robustness & Further Results

In this section we present results on spillovers, the extensive margin of reviews and neighborhood complaints. We then document multiple robustness checks.

6.1 Spillovers

The analysis is focused on tourists visiting a particular restaurant. We thus far have not tested if this effect spills over to restaurants located close by. We thus include in our baseline specification, Equation 1, measures of how many tourists visit restaurants in the surrounding area. As Table 6 shows, using different distances, we do not find strong evidence for that. The impact of a reduced influx of tourists is confined to the restaurant itself.

6.2 Geography of Consumption

Thus far, the analysis has focused on ratings and the review texts, and was thus conditional on observing a review. Next, we examine the extensive margin, i.e. whether we see a change in the geography of consumption. To do so, we estimate Equation 1, replacing the average rating with the number of reviews as dependent variable.[16] The hypothesis is that, when tourists are no longer around, locals start consuming at usually touristic venues.

The results in Table B.4 confirm our hypothesis. Returning to the example of mentioned above, the coefficient in column 2 implies that a restaurant near Notre-Dame cathedral is 3.9 percentage points more likely to be reviewed by a local in a given month (or 28.4 %), when tourists are not around. The improved experience evidenced by higher ratings is thus also reflected in the geography of consumption, with more touristic places receiving relatively more local visitors.

This effect does not seem to be driven by a broader shift across neighborhoods because

[16] As this is a (skewed) count variable containing zeros, we use both a linear probability model and a Poisson estimator to obtain our result. More precisely, we use a Poisson Pseudo Maximum Likelihood estimator.

of, for example, different commuting patterns. We find the effect to be robust to the inclusion of neighborhood-month fixed effects.[17]

6.3 Heterogeneity

We study heterogeneity along several neighborhood and restaurant characteristics. We find no heterogeneity across demographic and socio-economic neighborhood characteristics (see table C.9). Next, we show that restaurants in the largest mid-price range drive the results, rather than those at the tails of the price distribution (see table C.10. In fine dining, the luxurious context likely makes the additional presence of tourists less noticeable, while in cheap dining, the quick-service model may require little interaction, making tourist presence less relevant. Finally, we look at the linearity of our effect. Consistent with the descriptive evidence in Figure 3, we find that our result is driven by the top quartile most touristic restaurants. This suggests that some level of tourism can be sustained without observing the disamenity we describe.

6.4 Neighborhood Complaints

The main analysis above uses data coming from *Tripadvisor*. To provide further evidence that the lower influx of tourists improved locals' perceived satisfaction with local amenities, we analyze data on complaints registered within 100m of the restaurants in our sample by local residents (see Section 3 for a detailed description).

We estimate Equation 1, replacing the average rating of the restaurant with the number of complaints in the vicinity of a restaurant within a given month. Table B.7

[17]To further rule out a large shift in where locals consume, we compute the share of all reviews accounted for by each neighborhood, before and during the pandemic. In Figure A.8 we show that shares remain very similar, no broad shift in consumption has occurred.

presents the results. We find that complaints around touristic restaurants decline relative to less touristic ones. Using the estimate in column 2, the probability of registering a complaint around a restaurant near Notre-Dame cathedral in a given month decreases by around 2 percentage points (or 31 %).

The positive impact of a decrease in the arrival of tourists is thus not only reflected in restaurant ratings, but also confirmed by an entirely external data source, namely crowd-sourced complaints that are used to improve municipal services. Interestingly, the coefficient is reduced after including month-neighborhood fixed effects, indicating that some of the effect operates at a more aggregate level.[18]

6.5 Testing for Pre-Trends

In order to asses the timing of the effect that we find, we estimate Equation 1 allowing for β to be time-varying. In particular, we estimate one coefficient per quarter and set the first quarter of 2020 as reference group. If the effect is driven by the sudden and unexpected absence of tourists due to the pandemic, we should observe no differential trends for more touristic restaurants prior to the outbreak of COVID-19. Figure 5 shows that this is the case. Prior to the COVID-19 outbreak coefficients are close to and not statistically different from zero. Then, in Q3 and Q4 of 2020 coefficients are positive and statistically different from zero. This lends further support to the interpretation that COVID-19 led to a shift in locals' ratings of initially touristic venues. Similarly, Figure 7 shows the the absence of pretrends at the review level and Figure 6 for the November 2015 attacks.

[18]When we repeat the analysis at the neighborhood level we continue to find a strong reduction in complaints. This suggests that the disamenity caused by tourists at the restaurant level is mostly confined to the consumption experience. Other externalities of tourists spread more widely in the neighborhood.

6.6 Pedestrianized Streets

During the reopening of the hospitality sector after the first lockdown in France, one major concern was large indoor gatherings. As a countervailing measure, the mayor of Paris decided to pedestrianize some streets in the city.[19] Roads that were previously used by cars were now only accessible to foot traffic and the car lanes were partially occupied by restaurants.[20]To alleviate any concern that this measure disproportionately affected touristic restaurants (after conditioning on neighborhoods), we conduct a robustness exercise including a dummy if a restaurant is within a certain distance of one of these pedestrianized streets, interacted with a post-lockdown dummy, controlling for any differential trends among restaurants treated by this policy.[21] The results in Table C.6 show that our results remain unchanged.

6.7 Further Robustness Checks

We conduct numerous additional checks to which our results are robust (see Appendix C for details). First, we use a location-based tourism proxy instead of the language-based measure. Second, we vary the aggregation period used to construct the tourism proxy. Third, we allow for different trends depending on the pre-treatment number of reviews, as a proxy for restaurant capacity and popularity. Fourth, we account for the distance to the 2015 Paris attack sites. Fifth, we address data skewness by removing highly active users and popular restaurants. Sixth, we vary the level of clustering standard errors.

[19]See this Le Parisien article for a description of the pedestrianization of Paris during the summer of 2020, based on data from Marie de Paris: https://www.leparisien.fr/paris-75/bars-et-restaurants-a-paris-les-rues-qui-pourront-etre-occupees-par-des-terrasses-30-05-2020-8326956.php (last retrieved June 17, 2024)

[20]Figure C.1 shows a map of the targeted streets.

[21]We use the following distance cutoffs: 50, 100, 150 and 200 meters.

7 Mechanisms

To examine the mechanism, we look at the content of reviews. We rely on the text-based classification of reviews described in Section 3. This allows us to analyse, if reviews referring to certain topics are becoming more or less frequent. In particular, we estimate the following equation:

$$\text{Complaint on subject}_{ijt} = \beta \times \text{Post}_t \times \text{Tourism}_j + \gamma_j + \delta_t + \mu_i + \epsilon_{ijt}, \tag{3}$$

where *Complaint on subject*$_{jt}$ indicates whether review left by user i for restaurant j in month t referring to a particular subject of complaint, such as overcrowding. The dependent variable is measured in two different ways: it is a dummy for the dictionary-based approach and a continuous variable between 0 and 1 for the word embedding approach. The rest of the specification is as described in the section related to Equation 2. We also estimate a restaurant-level version of this specification.

We start by looking at supply-side changes. As tourism declines, restaurants may change what they offer. Since locals are more likely to be repeat customers, businesses may invest more in quality. Similarly, as demand declines, prices may decrease, increasing locals' satisfaction with amenities. As Columns 2 and 3 in Table 3 shows, we find no evidence for less complaints about prices or poor quality.

Next, we look at another common disamenity associated with tourism, overcrowding. A decrease of people in the city due to the decline in tourism may simply lead to less congestion. To get at this we look at comments mentioning either a long waiting time or a noisy environment. As Columns 4 and 5 in Table 3 show, we find no evidence pointing in this direction. More touristic restaurants did not receive relatively less reviews mentioning a long wait or noise after the lockdown. We interpret this as congestion not being a major driver of our results.

Overall, we thus find no evidence for disamenities typically associated with tourism. This is in line with the fact that we find no effect of the November 2015 attacks on ratings of tourists. Congestion, high prices, and, maybe to a lesser extent, changes in quality are likely to be negatively perceived by tourists, too. The absence of an effect on ratings by tourists makes these mechanisms unlikely to be at play. Instead, this suggests that the presence of tourists affects locals differently.

Since we find no evidence for more classical disamenities, we next look for social frictions, i.e. a direct, taste-based aversion of locals against tourists. As the first column in Table 3 shows, the only reviews that explicitly mention tourists appear significantly less after the lockdown in initially touristic places. This suggests that it is something about the presence of tourists themselves rather than perceived overcrowding or decreases in quality.[22] In Tables D.3, D.4, and D.5, we also verified the predictive relationship between the presence of different complaint types and ratings. Importantly, tourism-related complaints stood out, demonstrating significant explanatory power in our regressions. For example, including tourism-related complaints as controls in the differences-in-differences model, with ratings as the outcome, completely absorbs the tourism effect on ratings, thereby supporting our interpretation of social frictions as a mechanism.

To further test whether social frictions are at play, we exploit the composition of tourists, which varies by restaurant. We test whether the increase in ratings is higher when the tourists are socially more distant to the local population. In particular, we exploit the information on users' origin provided in their profile. This allows us to compute for each

[22]Table C.3 replicates this result using the location-based tourism measure.

restaurant the share of reviewers from a given country of origin. We combine this with the Social Connectedness Index (SCI) to compute the average SCI between restaurants' foreign reviewers and France.[23]

If Parisians have a distaste for foreigners from less familiar countries, we should see a higher increase in satisfaction for restaurants with many visitors from these countries. We thus estimate the treatment effect separately for restaurants with above and below-median SCI value. Table 4 shows that the increase in ratings of touristic places is indeed driven by low-SCI restaurants. For example, in Column 3, when including month-neighborhood fixed effects, the treatment effect for high-SCI is close to and not statistically different from zero. The coefficient for low-SCI places on the other hand suggests that touristic, low-SCI restaurants increased their rating.

This evidence is thus consistent with social frictions. Locals are less bothered by tourists who are similar to them.[24] We can also see in Table 5 that, in a similar way, restaurants frequented by tourists from countries with a lower SCI index show a higher likelihood of complaints related to the presence of tourism. However, we do not observe any statistically significant results for other subjects of complaints.

8 Conclusion

This paper studies the impact of tourism on locals' satisfaction with a key urban amenity, restaurants. We construct a granular dataset of urban consumption, which allows us

[23]See Section 3.5 for a description of the SCI.

[24]One concern might be that social connectedness is correlated with actual tourist arrivals from a country during the post-lockdown summer. However, the nature of the shock is such that arrivals from all countries drop to almost zero. Identification is thus almost entirely based on the pre-COVID exposure to tourism. In unreported results we control for differential changes in demand by nationality using a Bartik-style shock and find almost no change in our estimates.

to introduce a restaurant-level measure of tourism. Exploiting two different exogenous declines in tourism, we find robust evidence that the absence of tourists increases locals' satisfaction with restaurants.

Using recently developed text analysis methods to analyse the content of reviews, we find no evidence that the effect operates through a reduction in overcrowding or through changes in prices or quality. Instead, the results point towards social frictions. Reviews that explicitly relate to the presence of tourists become less frequent. In addition, the negative effect of tourism is driven by restaurants where the initial tourist clientele was from countries that have few social ties with the French population. Finally, tourists themselves are not affected by having less tourists around them, lending further support to the existence of social frictions between tourists and locals.

While there is evidence that tourism can have adverse economic effects on locals, e.g. by increasing rents, our results suggest that at least some of the widely spread attempts to limit tourism may be driven by hostility towards foreigners. The physical presence of tourists creates social frictions and decreases locals' perceived quality of life. This calls for a careful evaluation of seemingly rational arguments that are used to curb tourism. We leave it to future research to study determinants of the degree of social frictions and how policy makers can try to reduce them.

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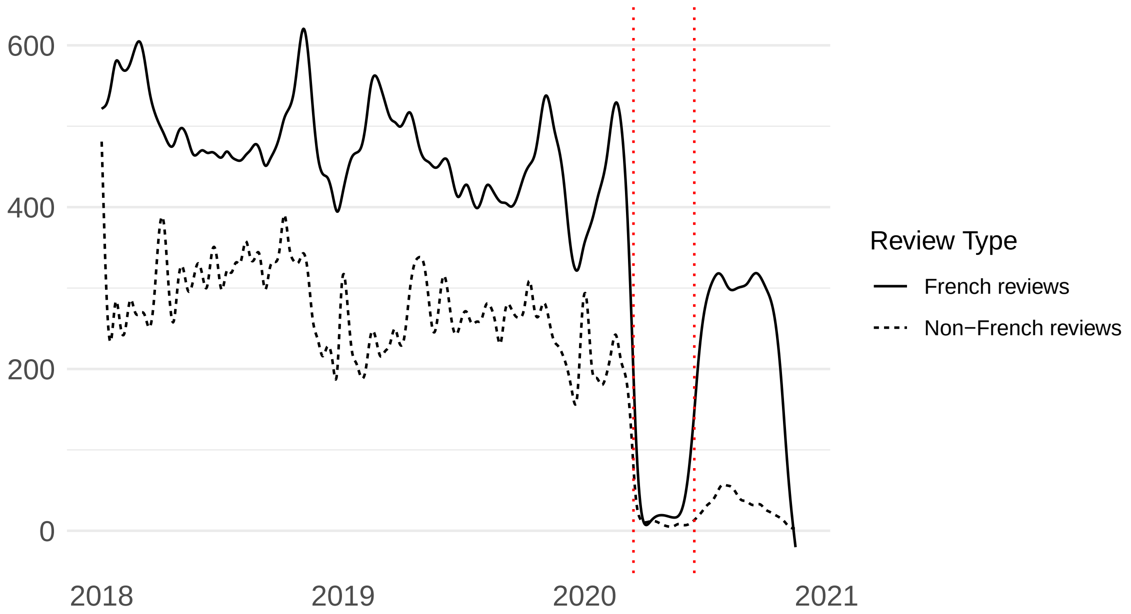
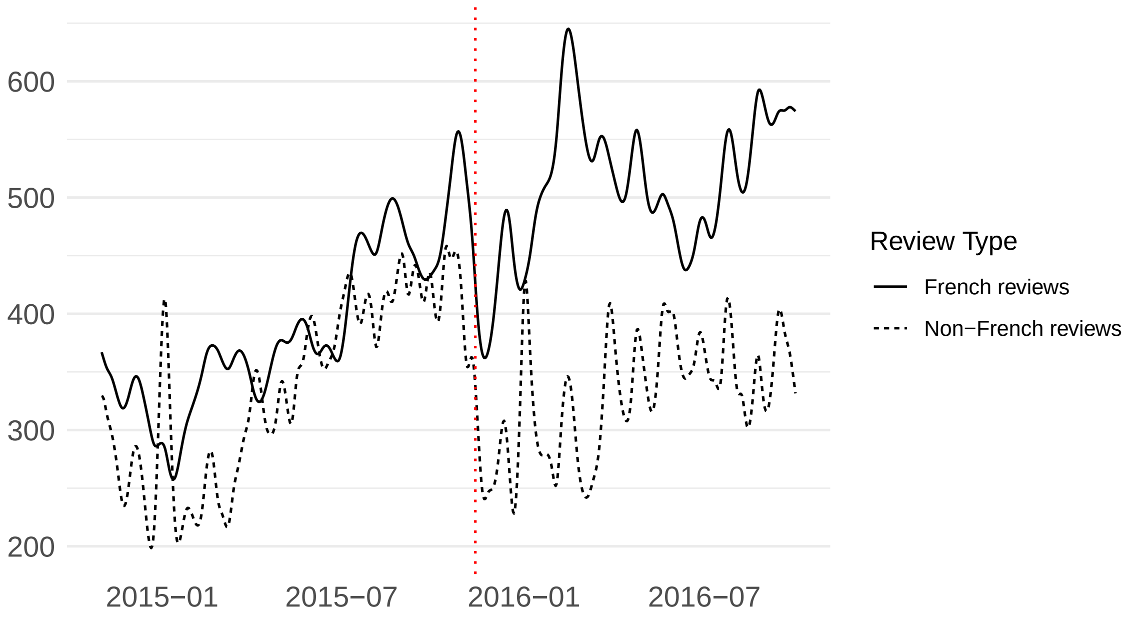
Figure 1: No Tourists in Summer 2020: French Restaurant Reviews Partially Recovered after the First Lockdown, while Reviews in Other Languages Remained Close to Zero



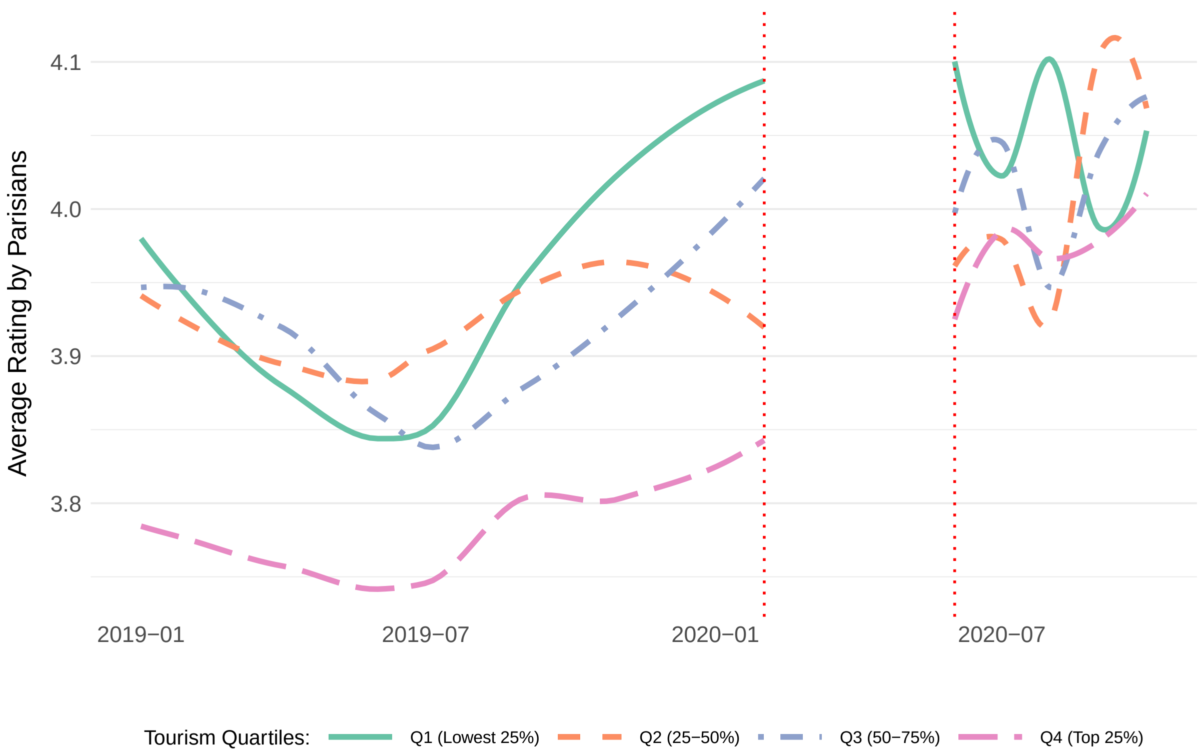
Notes: This figure illustrates the impact of the pandemic on restaurant reviews. Both lines correspond to the trends in the daily number of reviews, which are represented with smoothing splines, from January 2018 to November 2020. The solid line shows the French language reviews, while the dotted line represents non-French reviews. The two vertical red dotted lines mark the period of the first pandemic lockdown in Paris, which occurred between March and June 2020. During this period, restaurants were closed for visits. In the summer of 2020, the number of French reviews partially recovered, while non-French reviews remained close to zero.

Figure 2: After the November 2015 Terrorist Attack in Paris, the Number of Non-French Reviews Dropped



Notes: This figure illustrates the impact of the 2015 terrorist attacks on restaurant reviews. Both lines correspond to the trend in the daily number of reviews, which are represented with smoothing splines, from November 2014 to November 2016. The solid line shows the French language reviews, while the dotted line represents non-French reviews. The vertical red dotted line marks the day of the terrorist attack. After the attack, the number of French and non-French reviews diverged.

Figure 3: Evolution of Average Ratings by Parisians Across Different Percentiles of the Tourism Proxy

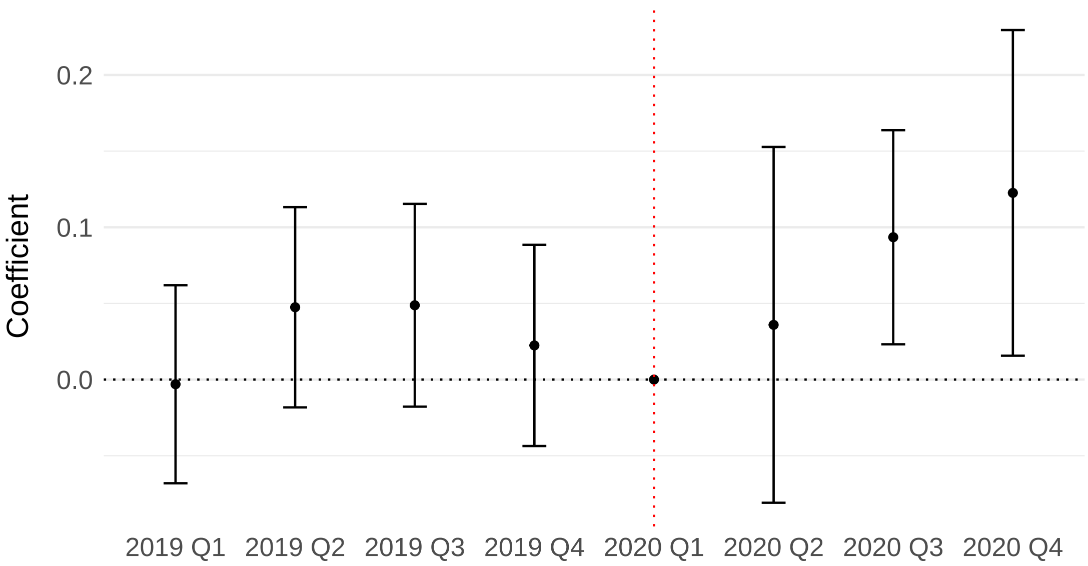


Notes: This figure illustrates the evolution of average restaurant ratings given by Parisians across different quartiles of the tourism proxy. The quartiles are based on the proportion of reviews written by non-French users before 2019, with Q1 representing the lowest 25% and Q4 representing the top 25%. The lines are smoothed using LOESS to highlight trends over time. The two vertical red dotted lines indicate the start and end of the first pandemic lockdown in Paris, from March to June 2020. During this period, restaurant activity was restricted, which influenced the observed trends in ratings.

Figure 4: The Map of Restaurants, Categorized by the Share of Non-French Reviews, Corresponds to the Distribution of Major Tourist Attractions

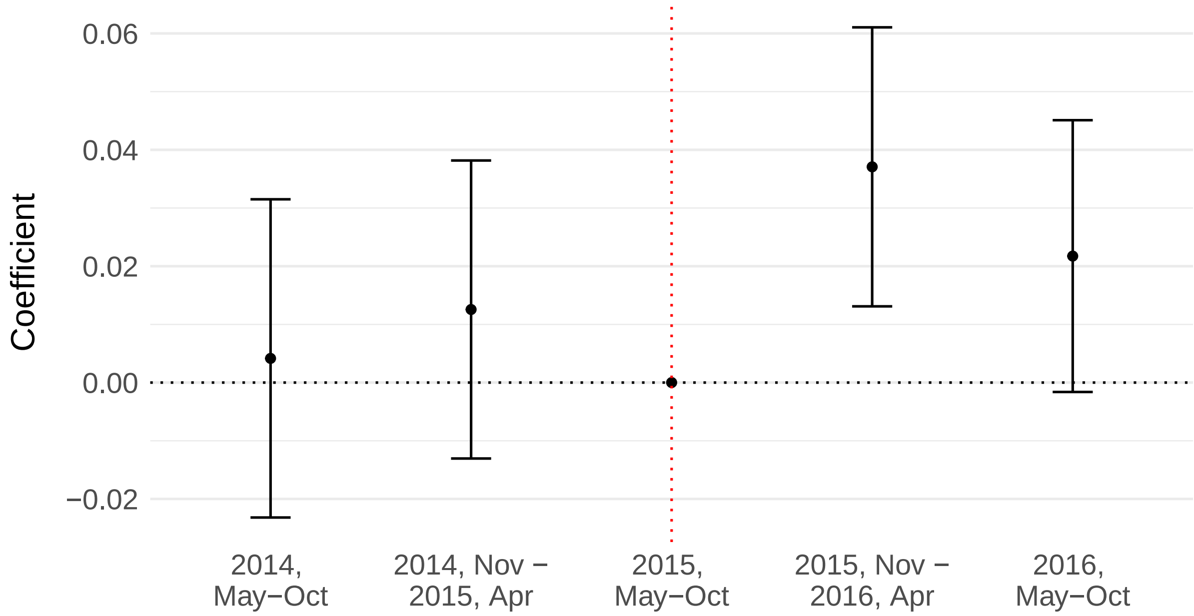
Notes: This map illustrates the distribution of restaurants based on the share of non-French reviews. Each point corresponds to a restaurant, and the color indicates the proportion of non-French reviews out of the total reviews prior to 2020. The color distribution indicates the level of tourist presence in the neighborhood, as the majority of restaurants with a high share of non-French reviews are clustered around major tourist attractions. The grid cell map, which mirrors the distribution of touristic restaurants, is depicted in Appendix Figure A.4.

Figure 5: Event Study Plot: Touristic Restaurants Have Relative Improvement in Ratings After Pandemic, Restaurant-Level Specification



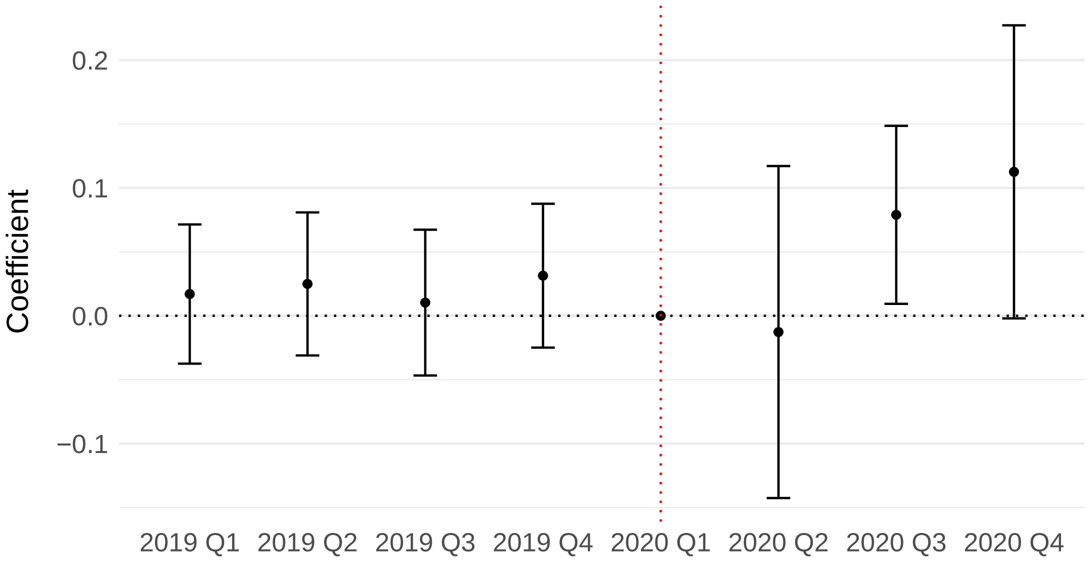
Notes: This figure displays the results of the event study concerning the pandemic-induced shock, where the unit of analysis is month × restaurant. Point estimates and 95% confidence intervals are provided. The first quarter of 2020 serves as the excluded time period. Standard errors are clustered at the quarter level.

Figure 6: Event Study Plot: Touristic Restaurants Have Relative Improvement in Ratings After November 2015 Attack, Restaurant-Level Specification



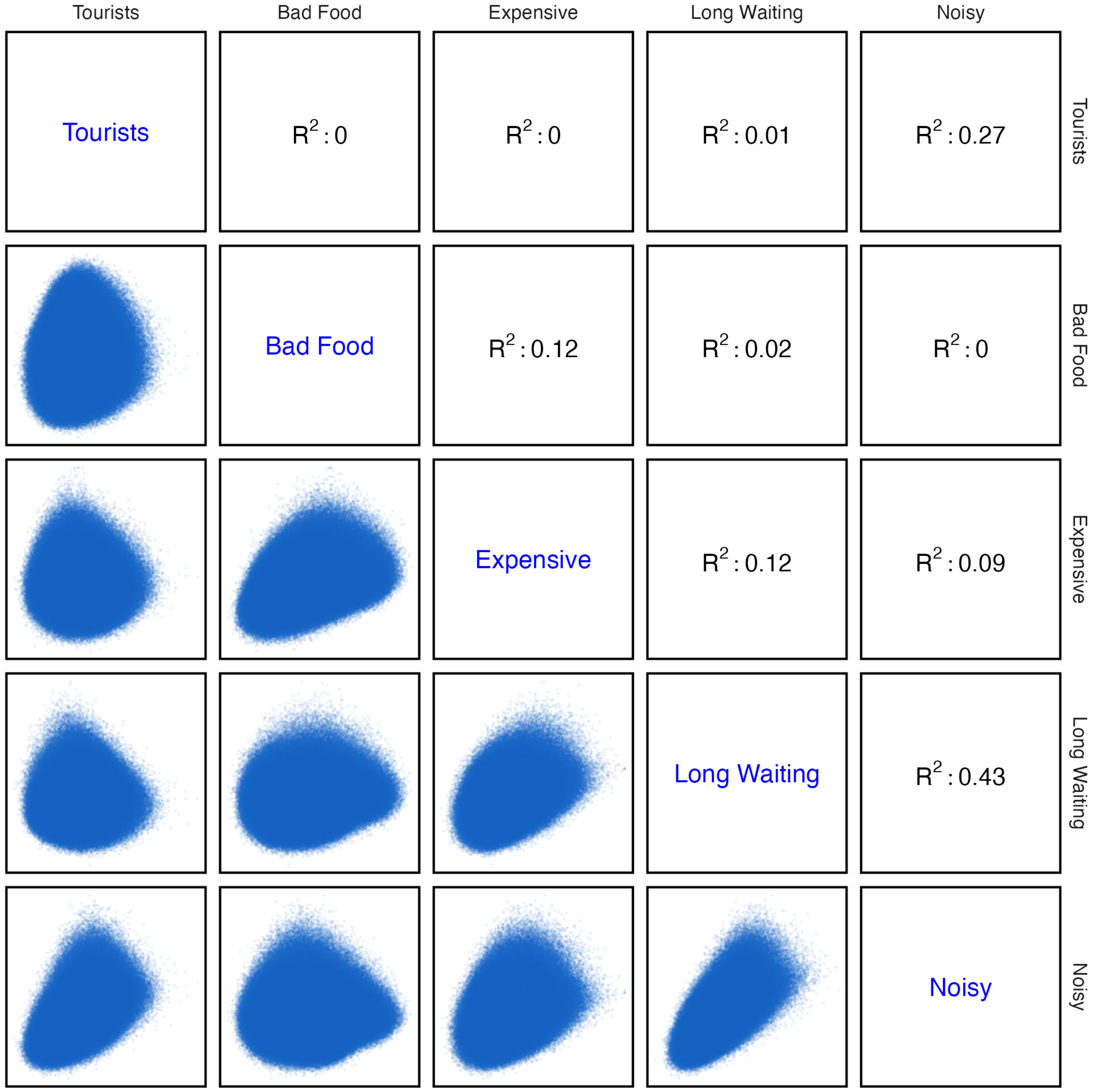
Notes: This figure displays the results of the event study concerning the terrorism-induced shock, where the unit of analysis is month × restaurant. Point estimates and 95% confidence intervals are provided. Points are grouped by half-a-year periods. The period from May to October 2015 serves as the excluded time period. The plot with quarter-level grouping is presented in Appendix Figure A.11. Standard errors are clustered at the quarter level.

Figure 7: Event Study Plot: Touristic Restaurants Have Relative Improvement in Ratings After Pandemic, Review-Level Specification



Note: This figure displays the results of the event study concerning the pandemic-induced shock, where the unit of analysis is review. Point estimates and 95% confidence intervals are provided. The first quarter of 2020 serves as the excluded time period. Standard errors are clustered at the quarter level.

Figure 8: Matrix of Correlations Between Five Types of Concerns in Reviews Measured as Cosine Distances in *word2vec*



Notes: This figure presents a correlation matrix for five distinct types of concerns found in review texts: (1) mentions of tourists, (2) poor food quality, (3) excessive prices, (4) long waiting times, and (5) excessive noise in the restaurant. Each variable is continuous and measured as the cosine distance between the centroid of words from a review and the centroid of a dictionary related to the subject, using a word2vec embedding trained on the primary corpus of reviews.

Table 1: Tourism Decreases Resident's Satisfaction with Urban Amenities: Restaurant Level Analysis, Difference-in-Differences

Natural experiments:	**Before and After First Pandemic Lockdown** *(Post = Post-Lockdown)*		**Before and After Terrorist Attack – November 2015** *(Post = Post-Terrorist Attack)*			
Dependent variables:	Avg. Rating by Parisians		Avg. Rating by Parisians		Avg. Rating by Non-Parisians	
	(1)	(2)	(3)	(4)	(5)	(6)
Share of Non-French Reviews prior to observation period (by Restaurant) × Post	0.0821***	0.0858***	0.0222**	0.0290**	-0.0036	-0.0101
	(0.0207)	(0.0255)	(0.0105)	(0.0120)	(0.0096)	(0.0102)
Fixed-effects						
Restaurant	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes		Yes		Yes	
Month × Neighborhood		Yes		Yes		Yes
Fit statistics						
Observations	72,471	72,471	42,033	42,033	42,033	42,033
R^2	0.36	0.38	0.36	0.39	0.37	0.40
Dependent variable mean	0.71	0.71	0.69	0.69	0.75	0.75
Dependent variable SD	0.31	0.31	0.27	0.27	0.21	0.21

Notes: This table presents OLS estimates. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 2: Tourism Decreases Resident's Satisfaction with Urban Amenities: Pandemic Shock, Review Level Analysis, Difference-in-Differences

Dependent variable:	Rating in Review Left by Parisian			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews (by Restaurant) × Post-Lockdown	0.0766***	0.0491*	0.0701**	0.0929**
	(0.0208)	(0.0259)	(0.0331)	(0.0413)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes	Yes		
User		Yes	Yes	
Month × Neighborhood			Yes	Yes
User × Post-Lockdown				Yes
Fit statistics				
Observations	112,238	112,238	112,238	112,238
R^2	0.29	0.74	0.75	0.77
Dependent variable mean	0.72	0.72	0.72	0.72
Dependent variable SD	0.33	0.33	0.33	0.33

Notes: This table presents OLS estimates. The unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-Lockdown* variable is a dummy variable, activated after the first pandemic lockdown. Standard errors are clustered at the neighborhood level. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 3: Drop in Tourism has Decreased Complaints about Tourists but has not Affected Other Types of Complaints: Pandemic Shock, Review-Level Analysis, Difference-in-Differences

Dependent variables: *(Topics of complaints)*	Tourists (1)	Low Food Quality (2)	Too Expensive (3)	Too Noisy (4)	Long Wait (5)
Panel A: Dictionary-Based (dummy := 1 if contains the keywords from the topic)					
Share of Non-French Reviews	-0.0919***	0.0040	-0.0328	0.0182	-0.0303
× Post-Lockdown	(0.0216)	(0.0308)	(0.0288)	(0.0268)	(0.0221)
Fixed-effects					
User × Post-Lockdown	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	112,238	112,238	112,238	112,238	112,238
R^2	0.57	0.61	0.54	0.48	0.54
Dependent variable mean	0.02	0.07	0.05	0.03	0.03
Dependent variable SD	0.15	0.26	0.22	0.17	0.16
Panel B: Word Embedding (Cosine Distances scaled to vary from 0 to 1)					
Share of Non-French Reviews	-0.0218**	-0.0074	-0.0120	-0.0096	-0.0060
× Post-Lockdown	(0.0094)	(0.0140)	(0.0093)	(0.0093)	(0.0100)
Fixed-effects					
User × Post-Lockdown	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	112,238	112,238	112,238	112,238	112,238
R^2	0.68	0.71	0.65	0.66	0.67
Dependent variable mean	0.51	0.53	0.51	0.50	0.50
Dependent variable SD	0.07	0.12	0.07	0.08	0.07

Notes: This table presents OLS estimates. The unit of analysis is a review. Panels A and B introduce two different measures of concerns expressed in reviews. In Panel A, the dependent variable is derived from the texts of reviews using dictionary-based method. In Panel B, the dependent variable represents the cosine distance between the centroid of words in reviews and the centroid of the dictionary related to a topic in word2vec embedding. The share of non-French reviews is calculated for the periods up to 2019. Post-lockdown is a dummy, which is switched on in June, 2020. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 4: Effect of Tourism on Residents' Satisfaction with Amenities Is Stronger for Restaurants Popular Among Tourists from Countries with Weaker Social Ties to France

Dependent variable:	Avg. Rating by Parisians			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews × Post-Lockdown	0.0738***	0.0748**		
	(0.0216)	(0.0306)		
Share of Non-French Reviews × Post-Lockdown × High Social Connectedness Index			0.0384	0.0295
			(0.0355)	(0.0419)
Share of Non-French Reviews × Post-Lockdown × Low Social Connectedness Index			0.0753***	0.0766**
			(0.0214)	(0.0303)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes		Yes	
Month x Neighborhood		Yes		Yes
Fit statistics				
Observations	59,275	59,275	59,275	59,275
R^2	0.34	0.37	0.34	0.37
Dependent variable mean	0.70	0.70	0.70	0.70
Dependent variable SD	0.31	0.31	0.31	0.31

Notes: This table presents OLS estimates. In all columns, the unit of analysis is the pair: month × restaurant. The dependent variable represents the average rating of restaurants among users with a home location in Paris. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-lockdown* variable is a dummy that is activated in June 2020, following the first COVID-19 lockdown. Measures of network proximity between countries of origin are constructed using Facebook data. Restaurants are categorized based on their proximity score into two groups: those above the median proximity (High SCI) and those below it (Low SCI). Standard-errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 5: Effect of Tourism on Complaints about Tourists is Higher for Restaurants Popular Among Tourists from Countries with Weaker Social Ties to France

Dependent variables: *(Topics of complaints)*	Tourists (1)	Low Food Quality (2)	Too Expensive (3)	Too Noisy (4)	Long Wait (5)
Share of Non-French Reviews × Post-Lockdown × High Social Connectedness Index	-0.0578***	0.0125	0.0452*	0.0117	0.0053
	(0.0162)	(0.0399)	(0.0268)	(0.0163)	(0.0178)
Share of Non-French Reviews × Post-Lockdown × Low Social Connectedness Index	-0.0811***	-0.0113	0.0132	0.0153	-0.0118
	(0.0153)	(0.0241)	(0.0185)	(0.0110)	(0.0136)
Fixed-effects					
Restaurant	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	59,275	59,275	59,275	59,275	59,275
R^2	0.24	0.22	0.19	0.18	0.19
Dependent variable mean	0.03	0.07	0.05	0.03	0.03
Dependent variable SD	0.15	0.24	0.20	0.14	0.14

Notes: This table presents OLS estimates. In all columns, the unit of analysis is the pair: month × restaurant. The dependent variable is derived from review texts using dictionaries and represents the share of reviews related to one of the corresponding topics, aggregated by restaurant-month. The share of non-French reviews is calculated for the periods up to 2019. The *Post-lockdown* variable is a dummy that activates in June 2020. Measures of network proximity between countries of origin are constructed using Facebook data. Restaurants are categorized based on their proximity scores into two groups: those with scores above the median (High SCI) and those below the median (Low SCI). Standard-errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 6: No Spillovers: Level of Tourism Around Restaurant Does Not Crowd Out Within Restaurant Effect

Dependent variables:	Avg. Rating by Parisians			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews × Post-Lockdown	0.0821***	0.0715***	0.0858***	0.0799***
	(0.0207)	(0.0238)	(0.0255)	(0.0249)
Touristic Area (<100m) × Post-Lockdown		-0.0105		0.0041
		(0.0282)		(0.0285)
Touristic Area (100m-300m) × Post-Lockdown		0.0824*		0.0690
		(0.0419)		(0.0488)
Touristic Area (300m-500m) × Post-Lockdown		-0.0377		-0.0303
		(0.0551)		(0.0521)
Touristic Area (500m-1000m) × Post-Lockdown		-0.0194		0.0487
		(0.0554)		(0.0919)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes	Yes		
Month × Neighborhood			Yes	Yes
Fit statistics				
Observations	72,471	72,471	72,471	72,471
R^2	0.36	0.36	0.38	0.38
Dependent variable mean	0.71	0.71	0.71	0.71
Dependent variable SD	0.31	0.31	0.31	0.31

Notes: This table reports OLS estimates. The unit of analysis is a pair month × restaurant. The dependent variable represents the average rating of restaurants given by users whose home location is Paris. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-lockdown* variable is a dummy that activates in June 2020. Standard errors clustered at the neighborhood level. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Online Appendix

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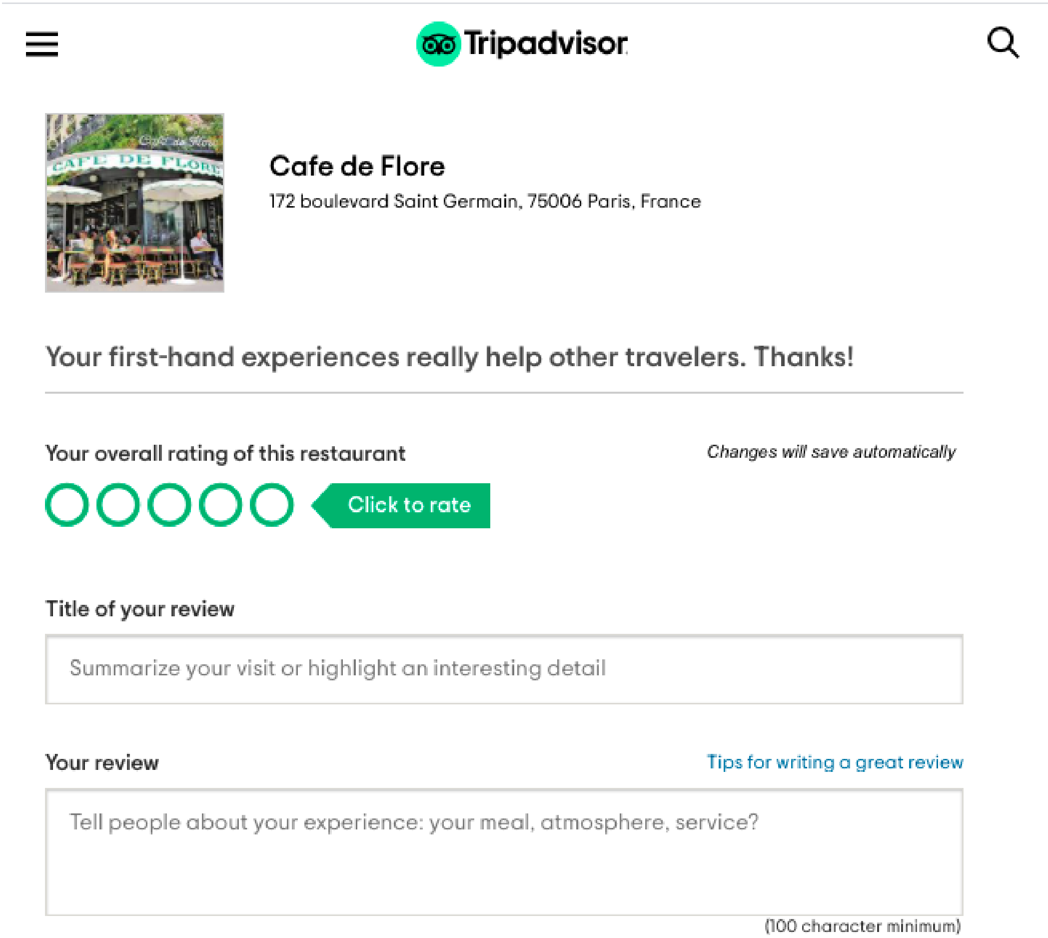
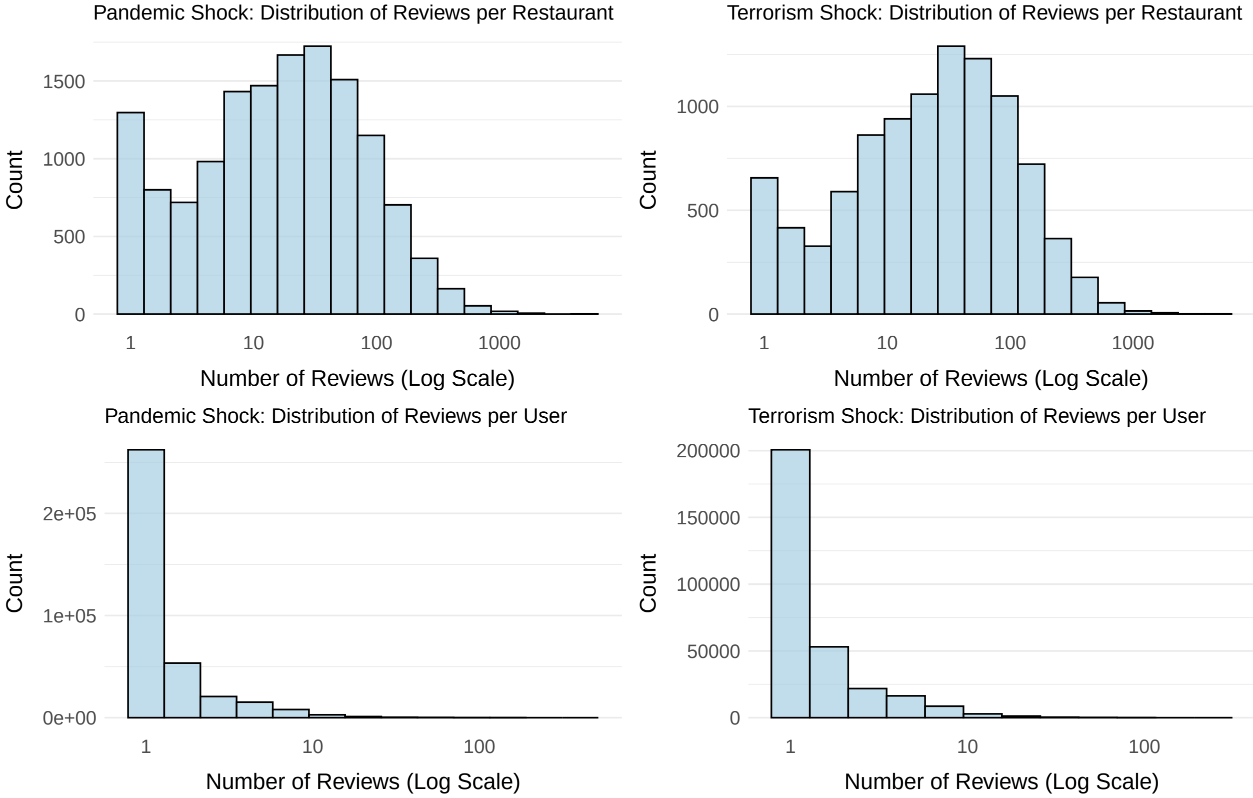
A Additional Plots

Figure A.1: Tripadvisor Interface



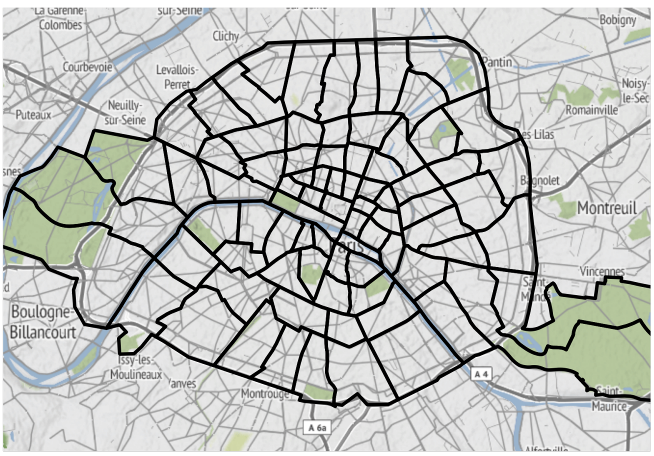
Notes: This figure is a screenshot that users of Tripadvisor see when they want to leave a review. *Café de Flore* is a famous Parisian café-restaurant in the Saint-Germain-des-Prés district, in the 6th arrondissement, popular among tourists and located in close proximity to the Sciences Po campus.

Figure A.2: Histograms of Review Counts by Shock Type (Pandemic vs. Terrorist Attack) and Level of Aggregation (Restaurants vs. Users)



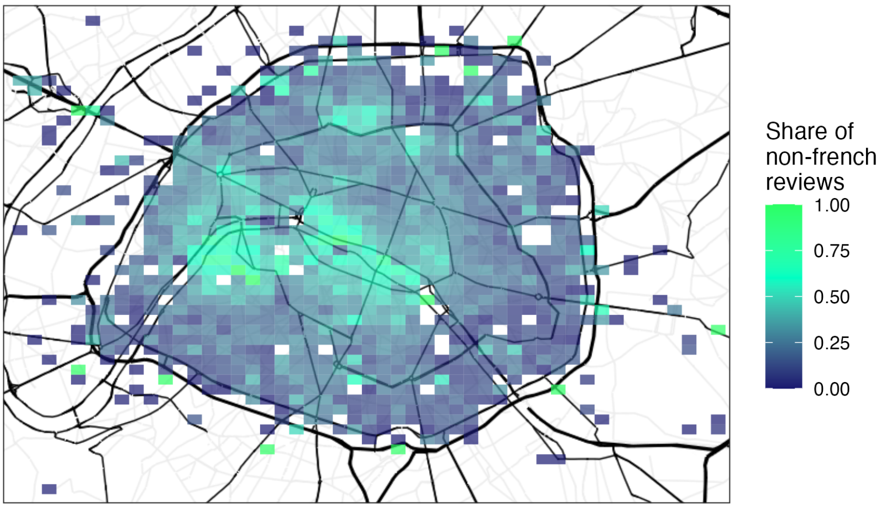
Notes: The figure presents the distribution of review counts at two levels of aggregation—restaurants and users—across two study periods: before and during the first wave of COVID-19 pandemic lockdowns, and before and after the Bataclan terrorist attacks. Each panel corresponds to a specific combination of study period and level of aggregation, with the x-axis showing the number of reviews on a logarithmic scale and the y-axis representing the frequency of observations.

Figure A.3: Parisian Neighbourhoods



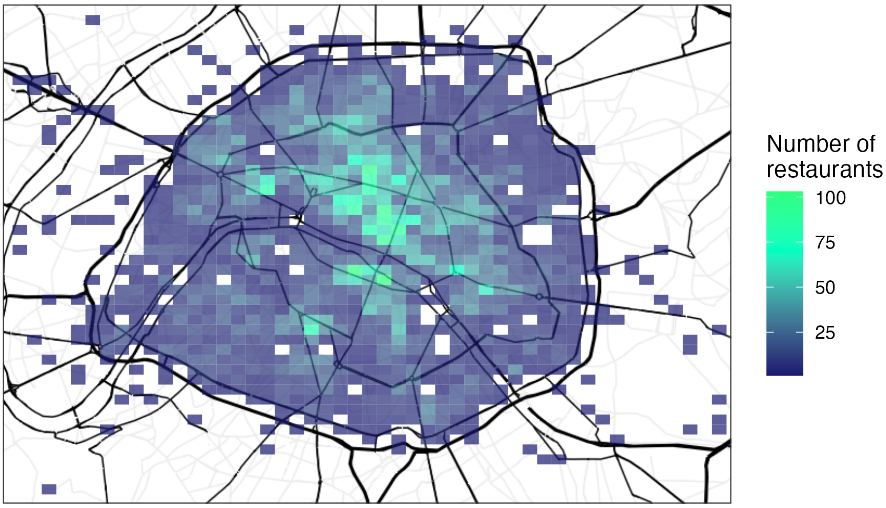
Notes: This map shows the definition of Paris' 80 administrative neighbourhoods, or "quartiers". The source is the Paris city hall.

Figure A.4: The Grid Map with the Shares of Non-French Reviews



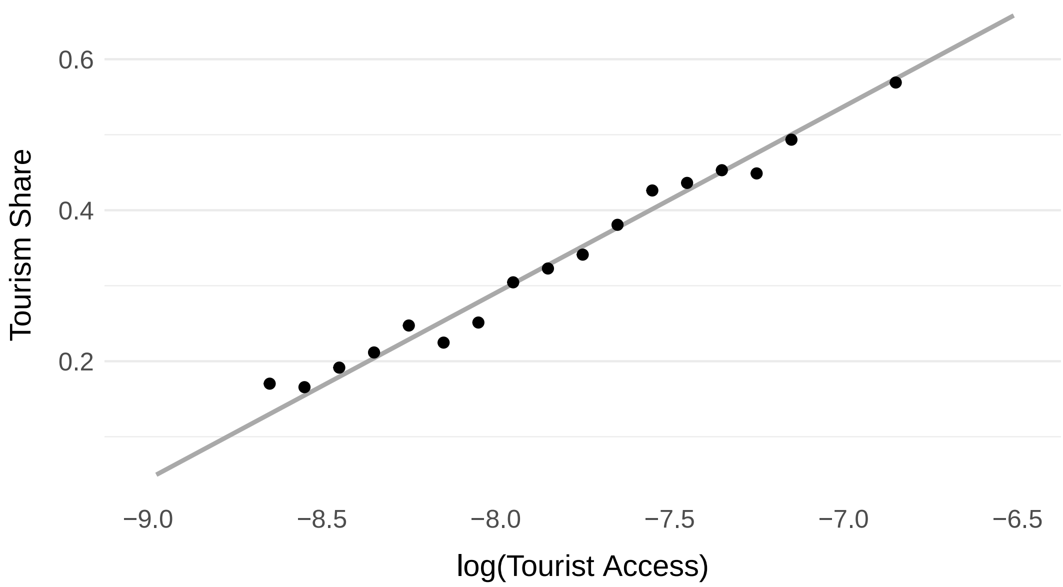
Notes: This map illustrates the distribution of restaurants based on the share of non-French reviews. Each gridcell's color corresponds to the proportion of non-French reviews out of the total reviews prior to 2020 in this area. The color distribution indicates the level of tourist presence in the neighborhood, as the majority of restaurants with a high share of non-French reviews are clustered around major tourist attractions.

Figure A.5: The Grid Map of Restaurant Density Differs from the Measure of Tourism Density



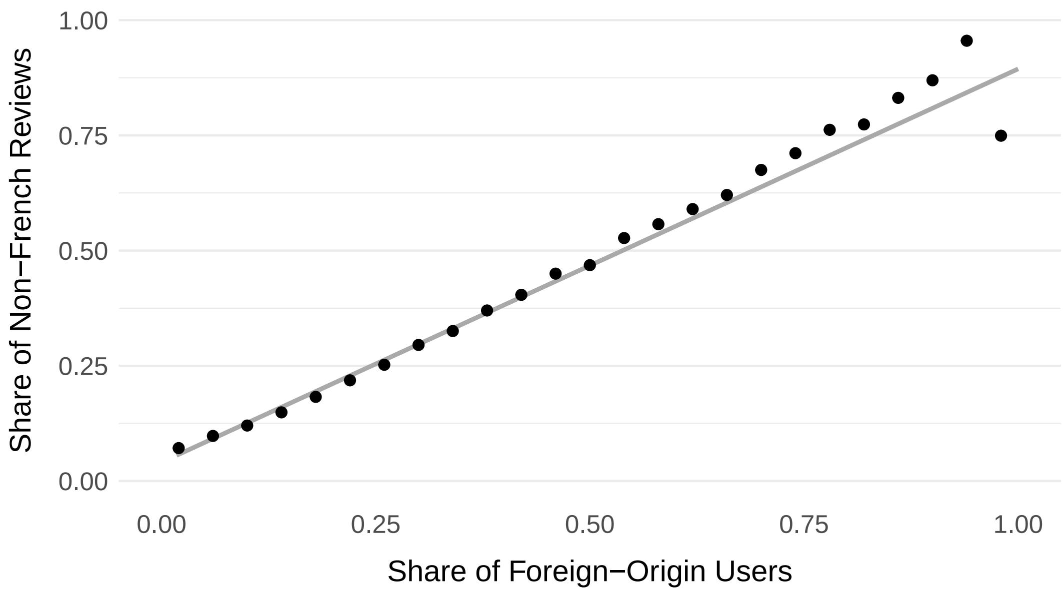
Notes: This map illustrates the density of restaurant locations. Each gridcell's color corresponds to the number restaurants prior to 2020 in this area. This color distribution indicates that the density of restaurants differs from the level of tourist presence in the neighborhood, as illustrated in Figure 4 and Appendix Figure A.4.

Figure A.6: Validation of Tripadvisor Tourism Measure with Tourist Access (Based on Number of Visitors to Major Attractions)



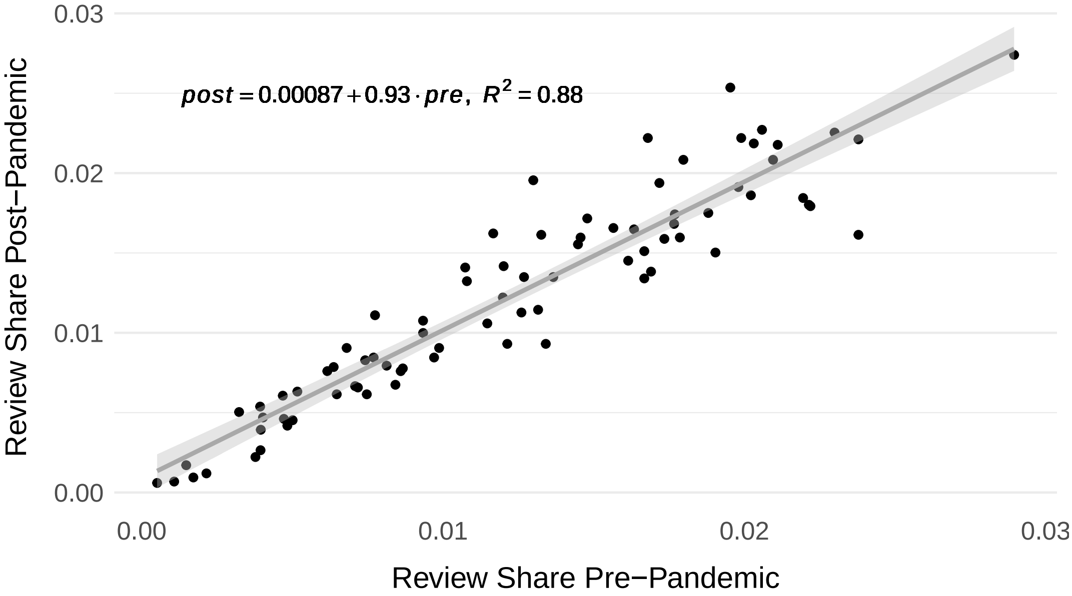
Notes: This binned scatter plot serves as a validation exercise for our baseline Tripadvisor measure of tourism, illustrating its correlation with the presence of tourism as measured by visitor numbers to major tourist attractions. The dataset captures the percentage of tourists in Paris visiting various attractions between 2015 and 2019. We geocoded 18 intra-muros attractions and then constructed a tourist demand measure using the market access framework.

Figure A.7: Validation of Language-Based Tourism Measure with Location-Based Tourism Measure



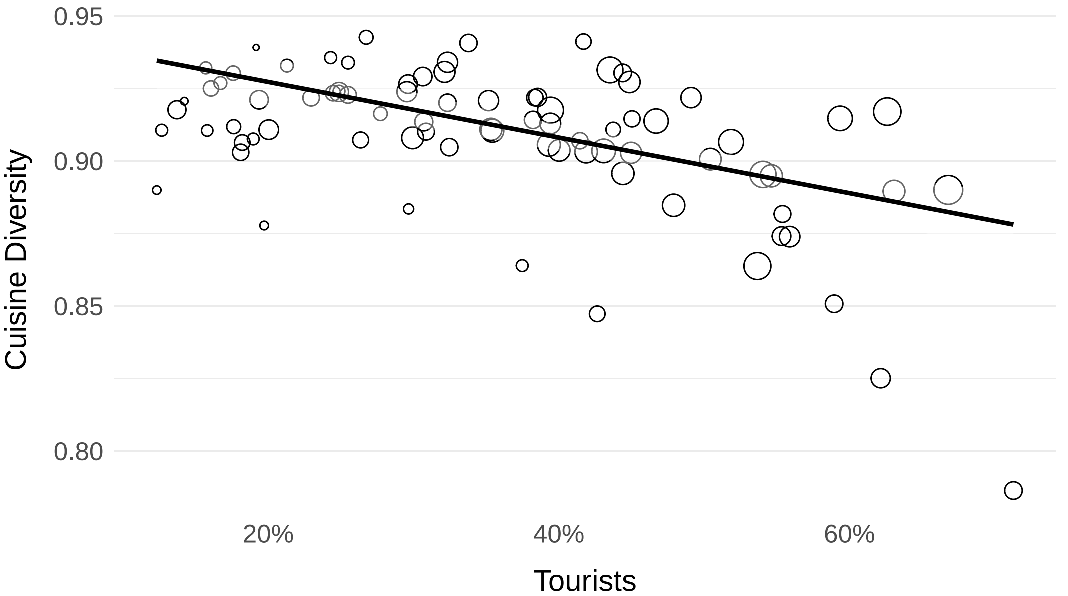
Notes: This figure is a binned scatter plot that illustrates the association between two measures of tourism based on TripAdvisor reviews: the share of non-French reviews and the share of reviews left by users with home location outside of France.

Figure A.8: Persistence of the Geography of Consumption



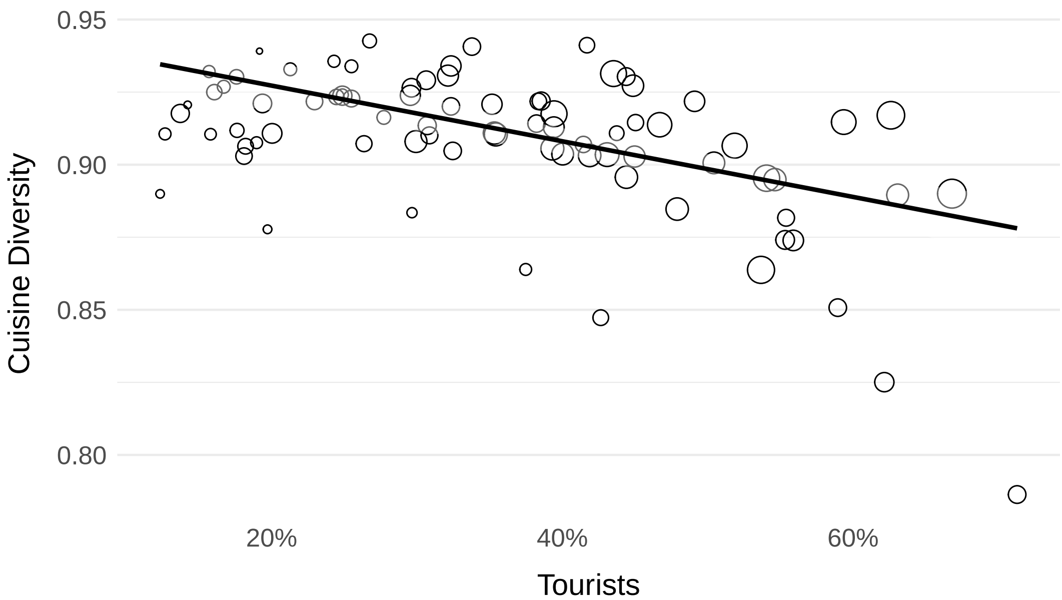
Notes: For this figure we compute the share of reviews accounted for by each of the 80 neighborhoods. We do this for the pre- and post-lockdown period. The figure shows the correlation.

Figure A.9: Share of French Cuisine and Tourism



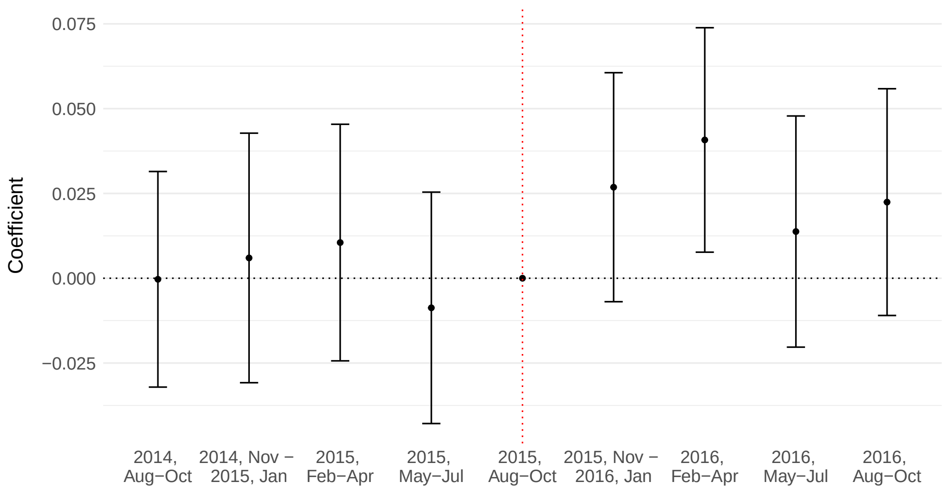
Notes: This figure demonstrates a positive correlation between the language-based measure of tourism and the share of restaurants serving French cuisine, both aggregated at the neighborhood level.

Figure A.10: Diversity of Cuisine Types and Tourism



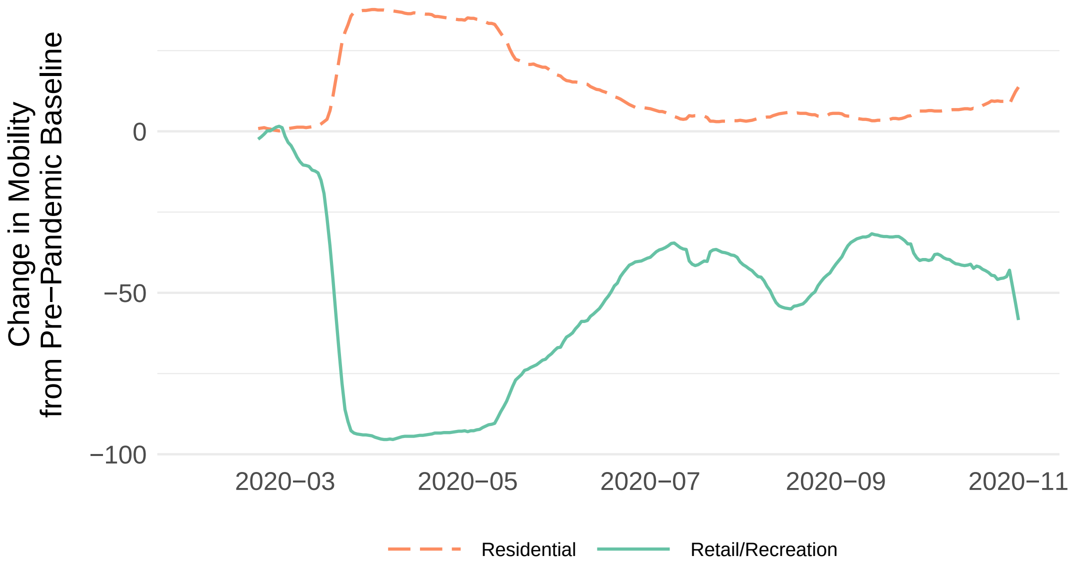
Notes: This figure shows a negative correlation between the language-based measure of tourism and cuisine diversity, with the latter measured as the reversed Herfindahl-Hirschman Index of cuisine types. Both metrics are aggregated at the neighborhood level.

Figure A.11: Event Study Plot: Touristic Restaurants Have Relative Improvement in Ratings After November 2015 Attack, Restaurant-Level Specification (Quarter Grouping)



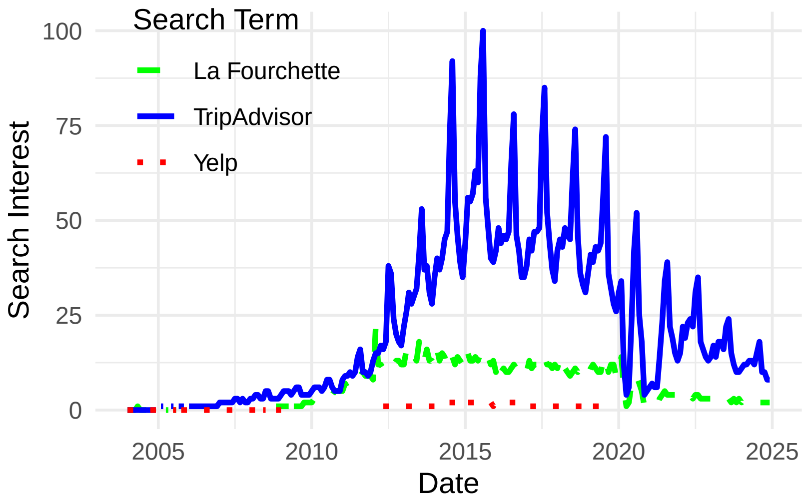
Notes: This figure displays the results of the event study concerning the terrorism-induced shock, where the unit of analysis is month × restaurant. Point estimates and 95% confidence intervals are provided. The period from August to October 2015 serves as the excluded time period. Standard errors are clustered at the quarter level.

Figure A.12: The Lockdown is Also Reflected in Mobility Patterns



Notes: This figure displays data from Google Mobility Trends, which relies on tracking the movements of users of Google's services. Google divides locations into multiple categories and provides data on the change in mobility relative to a baseline period from January to February 2020 for each of these categories. We plot retail/recreational space, the category closest to restaurants, and residential locations.

Figure A.13: TripAdvisor Outperforms Direct Competitors



Notes: This figure displays data from Google Trends, which relies on user queries in France from Google.

B Additional Tables

Table B.1: Descriptive Statistics, Main Analysis

	Unit of Observation	N	Mean	Median	Min	Q1	Q3	Max
Panel A: Pandemic Shock, Restaurant-Level Analysis								
Avg. Rating by Parisians	Restaurant-Month	72,471	3.86	4.00	1.00	3.00	5.00	5.00
Num. Reviews	Restaurant-Month	72,471	5.19	3.00	1.00	2.00	6.00	235.00
Tourists (Complaint)	Restaurant-Month	72,471	0.02	0.00	0.00	0.00	0.00	1.00
Food Quality (Complaint)	Restaurant-Month	72,471	0.07	0.00	0.00	0.00	0.00	1.00
Price (Complaint)	Restaurant-Month	72,471	0.05	0.00	0.00	0.00	0.00	1.00
Noise (Complaint)	Restaurant-Month	72,471	0.02	0.00	0.00	0.00	0.00	1.00
Long Wait (Complaint)	Restaurant-Month	72,471	0.03	0.00	0.00	0.00	0.00	1.00
Touristic Area (<100m)	Restaurant-Month	72,471	0.44	0.44	0.00	0.30	0.58	1.00
Touristic Area (100m-300m)	Restaurant-Month	72,471	0.46	0.46	0.00	0.35	0.56	1.00
Touristic Area (300m-500m)	Restaurant-Month	72,471	0.47	0.48	0.00	0.38	0.57	0.82
Touristic Area (500m-1000m)	Restaurant-Month	72,471	0.49	0.50	0.09	0.43	0.56	0.72
High Social Connectedness	Restaurant-Month	59,275	0.50	0.00	0.00	0.00	1.00	1.00
Non-French Reviews, Share	Restaurant	10,377	0.29	0.25	0.00	0.14	0.41	1.00
Panel B: Terrorist Attack Shock, Restaurant-Level Analysis								
Avg. Rating by Parisians	Restaurant-Month	42,033	3.77	4.00	1.00	3.00	4.60	5.00
Avg. Rating by Non-Parisians	Restaurant-Month	42,033	4.01	4.00	1.00	3.67	4.57	5.00
Num. Reviews	Restaurant-Month	42,033	8.23	6.00	2.00	4.00	9.00	281.00
Non-French Reviews, Share	Restaurant	5,281	0.41	0.39	0.00	0.21	0.60	1.00
Panel C: Pandemic Shock, Review-Level Analysis								
Rating by Parisian	Review	112,238	3.88	4.00	1.00	3.00	5.00	5.00
Tourists (Complaint)	Review	112,238	0.02	0.00	0.00	0.00	0.00	1.00
Food Quality (Complaint)	Review	112,238	0.07	0.00	0.00	0.00	0.00	1.00
Price (Complaint)	Review	112,238	0.05	0.00	0.00	0.00	0.00	1.00
Noise (Complaint)	Review	112,238	0.03	0.00	0.00	0.00	0.00	1.00
Long Wait (Complaint)	Review	112,238	0.03	0.00	0.00	0.00	0.00	1.00
Tourists (Distance)	Review	112,238	0.51	0.51	0.29	0.46	0.56	0.82
Food Quality (Distance)	Review	112,238	0.53	0.52	0.23	0.44	0.61	0.92
Price (Distance)	Review	112,238	0.51	0.50	0.30	0.46	0.55	0.89
Noise (Distance)	Review	112,238	0.50	0.50	0.29	0.44	0.55	0.88
Long Wait (Distance)	Review	112,238	0.50	0.49	0.31	0.45	0.54	0.83
Word Count	Review	112,238	64.22	49.00	9.00	32.00	78.00	1623.00
Non-French Reviews, Share	Restaurant	11,118	0.28	0.24	0.00	0.12	0.40	1.00

Notes: This table presents descriptive statistics for the variables used in the main analysis of the paper. Panel A summarizes variables for the restaurant-month level analysis examining the impact of the pandemic shock. Panel B focuses on the restaurant-month level analysis of the terrorist attack shock. Panel C provides review-level descriptive statistics for the pandemic shock.

Table B.2: Descriptive Statistics, User-Level

Statistic	N	Mean	Median	Min	Q1	Q3	Max	Period	Shock-Sample
Num. Reviews	44,719	2.45	1.00	1	1	2.00	321	Pre	Pandemic (All Users)
Propen. Visit Tourist	44,719	0.26	0.00	0	0	0.50	1	Pre	Pandemic (All Users)
Num. Reviews	7,023	1.59	1.00	1	1	1.00	52	Post	Pandemic (All Users)
Propen. Visit Tourist	7,023	0.26	0.00	0	0	0.50	1	Post	Pandemic (All Users)
Num. Reviews	3,031	7.09	3.00	1	1	8.00	321	Pre	Pandemic (Before&After Users)
Propen. Visit Tourist	3,031	0.26	0.17	0	0	0.40	1	Pre	Pandemic (Before&After Users)
Num. Reviews	3,031	2.04	1.00	1	1	2.00	54	Post	Pandemic (Before&After Users)
Propen. Visit Tourist	3,031	0.25	0.00	0	0	0.50	1	Post	Pandemic (Before&After Users)
Num. Reviews	144,963	1.73	1.00	1	1	2.00	87	Pre	Bataclan
Propen. Visit Tourist	144,963	0.40	0.00	0	0	1.00	1	Pre	Bataclan
Num. Reviews	25,899	2.16	1.00	1	1	2.00	186	Post	Bataclan
Propen. Visit Tourist	25,899	0.22	0.00	0	0	0.33	1	Post	Bataclan

Notes: The table presents descriptive statistics at the user level for various samples, broken down into pre- and post-treatment periods. For each Period-Shock combination, we report two types of statistics: the number of reviews per user and the propensity to visit restaurants in the top 25th percentile based on our tourism proxy. For the pandemic shock, we include two versions of the sample, corresponding to the two designs we employed: one comprising all users and another focusing on users who left reviews both before and after the shock.

Table B.3: Age Distribution Comparison

Age Group	**TripAdvisor Users**	**Parisians**
18–24	11.75%	–
25–34	28.09%	–
35–44	20.35%	–
45–54	17.91%	–
55–64	13.82%	–
65+	8.08%	–
15–29	–	23.9%
30–44	–	21.8%
45–59	–	18.5%
60–74	–	14.4%
75+	–	8.0%
Mean Age	≈ 41.5	≈ 39

Notes: This table presents demographic distributions for the Parisian population and TripAdvisor users. The data for Parisians is taken from the French National Institute of Statistics and Economic Studies, and the data for TripAdvisor users is taken from SimilarWeb. Overall, we observe that the demographic profile of TripAdvisor users is not drastically different from the average Parisian.

Table B.4: Effect on the Number of Reviews Left by Parisians

Dependent Variables:	Has Review by Paris		Number of Parisian Reviews		Has Review by Paris		Number of Parisian Reviews	
	OLS	OLS	Poisson	Poisson	OLS	OLS	Poisson	Poisson
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Share of Non-French Reviews × Post-Lockdown	0.0370***	0.0606***	0.4153***	0.5701***				
	(0.0057)	(0.0055)	(0.0887)	(0.0966)				
Share of Non-French Reviews × Post-Attack					0.0249***	0.0279***	0.2551***	0.2736***
					(0.0058)	(0.0059)	(0.0379)	(0.0416)
Fixed-effects								
Restaurant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes		Yes		Yes		Yes	
Month × Neighborhood		Yes		Yes		Yes		Yes
Fit statistics								
Observations	411,945	411,945	314,627	314,491	175,394	175,394	152,448	152,428
Dependent variable mean	0.19	0.19	0.38	0.38	0.33	0.33	0.69	0.69
Dependent variable SD	0.39	0.39	0.90	0.90	0.47	0.47	1.30	1.30

Notes: This table presents OLS and PPML estimates. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-2 and 5-6 is whether there is at least one review left by a Parisian during a given month in this restaurant. Columns 3-4 and 7-8 correspond to the total number of reviews left by Parisians. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic- and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table B.5: Non-Linear Effect on Restaurant Ratings

Dependent Variable:	Avg. Rating by Parisians				
	(1)	(2)	(3)	(4)	(5)
Post-Lockdown × Tourism $[25-50Q]$	0.0060				
	(0.0115)				
Post-Lockdown × Tourism $[50-75Q]$	0.0176				
	(0.0129)				
Post-Lockdown × Tourism $[75-100Q]$	0.0354**				
	(0.0142)				
Post-Lockdown × Tourism $[>25Q]$		0.0180			
		(0.0112)			
Post-Lockdown × Tourism $[>50Q]$			0.0211**		
			(0.0087)		
Post-Lockdown × Tourism $[>75Q]$				0.0259**	
				(0.0105)	
Post-Lockdown × Tourism $[>90Q]$					0.0526***
					(0.0132)
Fixed-effects					
Restaurant	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	72,471	72,471	72,471	72,471	72,471
R^2	0.36	0.36	0.36	0.36	0.36
Dependent variable mean	0.71	0.71	0.71	0.71	0.71
Dependent variable SD	0.31	0.31	0.31	0.31	0.31

Notes: This table presents OLS estimates. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for a single shocks in tourism: the first year of the pandemic. The dependent variable is the average rating of restaurants among users with a home location in Paris. All ratings are scaled from 0 to 1. The *Post-Lockdown* variable is a dummy, activated after the shock. We interact *Post-Lockdown* with various measures of tourism, based on the position of the corresponding restaurant in the overall distribution of restaurants by our measure of tourism presence.. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table B.6: Tourism Decreases Resident's Satisfaction with Urban Amenities: November 2015 Shock, Review Level Analysis, Difference-in-Differences

Dependent variables:	Rating in Review Left by Parisian			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews (by Restaurant) × Post-Attack	0.0287*** (0.0083)	0.0124 (0.0115)	0.0073 (0.0147)	0.0073 (0.0167)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes	Yes		
User		Yes	Yes	
Month × Neighborhood			Yes	Yes
User × Post-Attack				Yes
Fit statistics				
Observations	97,462	97,462	97,462	97,462
R^2	0.25	0.71	0.72	0.76
Dependent variable mean	0.70	0.70	0.70	0.70
Dependent variable SD	0.31	0.31	0.31	0.31

Notes: This table presents OLS estimates. In all columns, the unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2014. The *Post-Attack* variable is a dummy variable, activated after the November 2015 terrorist attack. Standard errors, clustered at the neighborhood level, are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table B.7: Tourism and "DansMaRue" Complaints

Dependent Variable:	Number of Complaints				
Level of Analysis:	*Restaurant*				*Neighborhood*
	OLS	OLS	Poisson	Poisson	Poisson
	(1)	(2)	(3)	(4)	(5)
Share of Non-French Reviews × Post-Lockdown	-0.0699**	-0.0313*	-0.6255***	-0.2377*	-0.2365**
	(0.0302)	(0.0158)	(0.2229)	(0.1338)	(0.1192)
Fixed-effects					
Restaurant	Yes	Yes	Yes	Yes	
Month	Yes		Yes		Yes
Month × Neighborhood		Yes		Yes	
Neighborhood					Yes
Fit statistics					
Observations	395,932	395,932	388,213	324,694	2,480
Dependent variable mean	0.18	0.18	0.90	1.1	484.8
Dependent variable SD	0.38	0.38	3.74	4.07	434.2

Notes: This table reports OLS and PPML estimates. The dependent variable is the number of complaints registered on the "Dans ma rue" platform within 100m of a restaurant in a given month. The share of non-French reviews is calculated for the periods up to 2019. Postlockdown is a dummy, which is switched on in June, 2020. Column 5 presents neighborhood-level analysis. Standard errors clustered at neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

C Robustness Checks

Below we discuss a series of robustness checks of our main result.

Location-based Tourism Proxy. Throughout our main analysis, we rely on a language-based measure to construct a proxy for tourism exposure for all restaurants in our dataset. However, for a subsample of users, a more refined measure is available, as their specified home location is known. Below, we present a series of robustness checks using this alternative location-based tourism proxy, demonstrating that our main findings remain robust. Table C.2 documents results for the restaurant × month and review-level design, and Table C.3 presents text analysis results using the location-based measure instead of the language-based measure.

Tourism Proxy Aggregated by Different Periods. To construct the tourism proxy used for treatment assignment, we calculate the share of non-French reviews for each restaurant during the pre-treatment period. In this robustness check, we show that our findings are not sensitive to the chosen aggregation period. Table C.4 presents results where we vary the cutoff year for included reviews in the tourism proxy, showing consistent outcomes.

Controlling for Pre-Treatment Number of Reviews. Our differences-in-differences approach assumes that the shocks we study affect tourist and non-tourist restaurants differently only through the absence or reduction of tourists. However, general conditions for restaurants changed following both the first pandemic lockdowns and the terrorist attacks in Paris. In some cases, our treatment assignment may be confounded by other unobserved trends. In this robustness check, we control for the pre-treatment total number of reviews as a proxy for both the capacity and popularity of restaurants. Specifically, we include this factor as an interaction term, Number of Reviews × Post, to account for potential differences. The rationale is that tourist-oriented restaurants generally have higher capacity and past customer flow, which may enable them to adapt to sanitary requirements more easily. Table C.5 shows that including this factor does not affect our findings.

Streets Pedestrianized during Summer 2020. As previously mentioned, during the summer of 2020, Parisian restaurants were open and subsidized by the state. However, not everything remained unchanged. One significant innovation during this time was the pedestrianization of some streets.[25] This change could have differentially affected tourist and non-tourist restaurants, potentially influencing our results and thus compromising the causal interpretation of the link between tourism and satisfaction with amenities.

To address this concern, we identified all newly pedestrianized streets from the summer of 2020 and created buffers around these streets at 50, 100, 150, and 200 meters (see Figure C.1 for illustration). We then added a new control to our main specification: an

[25] See this Le Parisien article for a description of the pedestrianization of Paris during the summer of 2020, based on data from Mairie de Paris: https://www.leparisien.fr/paris-75/bars-et-restaurants-a-paris-les-rues-qui-pourront-etre-occupees-par-des-terrasses-30-05-2020-8326956.php (last retrieved June 17, 2024)

interaction between a dummy variable indicating whether the restaurant falls within the buffers and a dummy for before and after the shock. Table C.6 shows that our main results are not sensitive to the inclusion of this variable.

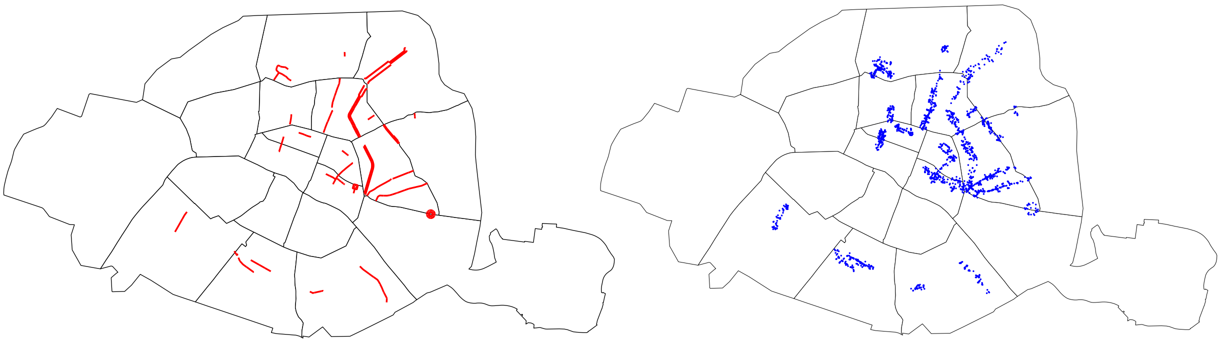
Controlling for Distance to Places of Attack. The November 2015 terrorist attacks in Paris unfolded simultaneously at multiple locations across the city, including near Stade de France, Rues Bichat and Alibert (Le Petit Cambodge; Le Carillon), Rue de la Fontaine-au-Roi (Café Bonne Bière; La Casa Nostra), the Bataclan theatre, Rue de Charonne (La Belle Équipe), and Boulevard Voltaire. This analysis examines the sensitivity of our key findings to distances from these attack sites to determine if our results were influenced by fear or increased support for the affected areas. The inclusion of controls for proximity to these locations (interacted with a post-attack dummy) would impact the coefficient associated with our tourism proxy if this were the case. However, Table C.7 demonstrates that this is not the case, as the main result remains unchanged.

Accounting for Skewness. User-generated platform and social media data is often skewed by a small number of highly active users or accounts, and restaurant review data is no exception. In this exercise, we address this issue by removing the most active users and the most popular restaurants from the sample. Table C.8 shows that our findings remain robust to these sample restrictions.

Heterogeneity. To further investigate our main results and verify whether they are indeed driven by the relative absence of tourists post-treatment rather than other channels, we conduct heterogeneity analyses. Table C.9 examines interactions with neighborhood census-based characteristics, such as the average demographic profile and socio-economic status. The results indicate that this heterogeneity does not significantly impact our findings. Additionally, we explore heterogeneity by restaurant price category, as defined on the platform (cheap dining, moderate dining, and fine dining). Table C.10 shows that our results are primarily driven by the most numerous middle category, which aligns with our interpretation. In fine dining, the structured and organized nature of operations makes the additional presence of tourists less noticeable, while in cheap dining, the quick-service model often does not require on-site interaction, making tourist presence less relevant.

Clustering. In Table C.11, we present our estimates using alternative methods of clustering standard errors, finding that, in all cases, our main results remain statistically significant.

Figure C.1: Pedestrianized Streets, Buffers and Restaurants within the Buffers



Notes: This map illustrates the streets that were pedestrianized during the summer of 2020. The panel on the left highlights the streets and squares. The panel on the right shows the restaurants within 100-meter buffers near this list. The source is from the Mairie de Paris. This is the full list of the streets and squares: *Rue des Jeuneurs*, *Quartier Sainte-Anne*, *Quartier du Carreau-du-Temple*, *Quartier du Marais*, *Place des Vosges et rue de Birague*, *Sortie du métro Le Pelletier*, *Quartier du canal Saint-Martin*, *Quartier du Faubourg-St-Denis*, *Quartier Sainte-Marthe*, *Bd de Belleville*, *Bd Richard-Lenoir*, *Rue de Charonne et rue de la Roquette*, *Place de la Nation*, *Rue du Chevaleret*, *Quartier Butte-aux-Cailles*, *Rue Daguerre*, *Rue du Château*, *Rue du Commerce*, *Rue Lepic*, *Rue des Abbesses*, *Quartier du marché de l'Olive*, *Canal de l'Ourcq*, *Rue Jourdain*

.

Table C.1: Tourist Access

	Tourism Share			
	(1)	(2)	(3)	(4)
log(Tourist Access)	0.2260***	0.2014***	0.2358***	0.1145***
	(0.0170)	(0.0435)	(0.0279)	(0.0373)
Weighted			Yes	Yes
Fixed-effects				
Neighborhood		Yes		Yes
Fit statistics				
Observations	13,908	13,908	13,908	13,908
R^2	0.16	0.23	0.25	0.39
Dependent variable mean	0.30	0.30	0.30	0.30

Notes: This table presents OLS estimates. It demonstrates a validation exercise for our baseline measure of tourism, illustrating its correlation with the presence of tourism as measured by visitor numbers to major tourist attractions. The dataset captures the percentage of tourists in Paris visiting various attractions between 2015 and 2019. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.2: Location-Based Measure: Tourism and Restaurant Ratings by Parisians

Dependent variables:	Rating by Parisians					
	Restaurant-Level		User-Level			
	(1)	(2)	(3)	(4)	(5)	(6)
Share of Non-Parisians Reviews × Post-Lockdown	0.1150*** (0.0233)	0.1017*** (0.0254)	0.1117*** (0.0250)	0.0785** (0.0299)	0.0879*** (0.0322)	0.0954** (0.0372)
Fixed-effects						
Restaurant	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes		Yes	Yes		
Restaurant-Month		Yes			Yes	Yes
User				Yes	Yes	
User-Post						Yes
Fit statistics						
Observations	72,424	72,424	112,176	112,176	112,176	112,176
R^2	0.36	0.38	0.29	0.74	0.75	0.77
Dependent variable mean	0.71	0.71	0.72	0.72	0.72	0.72

Notes: This table presents OLS estimates. In the first two columns the unit of analysis is a month × restaurant pair. The dependent variable is the average rating of restaurants among users with a home location in Paris. In the third to sixth column the unit of analysis is a review. All ratings are scaled from 0 to 1. The share of non-Parisian reviews is calculated for the periods up to 2019. The *Post-Lockdown* variable is a dummy, activated after the first pandemic lockdown. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.3: Location-Based Measure: Textual Outcomes

Dependent variables:	Tourists (1)	Low Food Quality (2)	Too Expensive (3)	Too Noisy (4)	Long Wait (5)
Share of Non-Parisians	-0.0778***	-0.0266	-0.0433	0.0245	-0.0082
Reviews × Post-Lockdown	(0.0201)	(0.0404)	(0.0328)	(0.0257)	(0.0267)
Fixed-effects					
User × Post-Lockdown	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	112,176	112,176	112,176	112,176	112,176
R^2	0.57	0.61	0.54	0.48	0.54
Dependent variable mean	0.02	0.07	0.05	0.03	0.03

Notes: This table presents OLS estimates. In all columns, the unit of analysis is a review. The dependent variable is derived from review texts using dictionaries and indicates if reviews are related to one of the corresponding topics. The share of non-Parisian reviews is calculated for the periods up to 2019. The *Post-lockdown* variable is a dummy that activates in June 2020. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.4: Tourism and Ratings: Language-Based Tourism Aggregated by Different Periods

Dependent variables:	Avg. Rating by Parisians				
	(1)	(2)	(3)	(4)	(5)
Share of Non-French Reviews (< 2017) × Post-Lockdown	0.0653**				
	(0.0284)				
Share of Non-French Reviews (< 2018) × Post-Lockdown		0.0801***			
		(0.0274)			
Share of Non-French Reviews (< 2019) × Post-Lockdown			0.0882***		
			(0.0235)		
Share of Non-French Reviews (< 2020) × Post-Lockdown				0.0858***	
				(0.0255)	
Share of Non-French Reviews (< 2021) × Post-Lockdown					0.0853***
					(0.0275)
Fixed-effects					
Restaurant	Yes	Yes	Yes	Yes	Yes
Month x Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	54,639	62,524	68,907	72,471	72,471
R^2	0.38	0.37	0.38	0.38	0.38
Dependent variable mean	0.70	0.70	0.71	0.71	0.71

Notes: This table presents OLS estimates. In all columns, the unit of analysis is a month × restaurant pair. The dependent is the average rating of restaurants among users with a home location in Paris. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to a specified year. The *Post-lockdown* variable is a dummy, activated after the first pandemic lockdown. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.5: Controlling for the Pre-Treatment Number of Reviews

Natural experiments:	**Before and After First Pandemic Lockdown** *(Post = Post-Lockdown)*			**Before and After Terrorist Attack – November 2015** *(Post = Post-Terrorist Attack)*			
Dependent variables:	Avg. Rating by Parisians			Avg. Rating by Parisians		Avg. Rating by Non-Parisians	
	(1)	(2)		(3)	(4)	(5)	(6)
Share of Non-French Reviews prior to observation period (by Restaurant) × Post	0.0780***	0.0829***		0.0256**	0.0313**	-0.0028	-0.0084
	(0.0216)	(0.0266)		(0.0114)	(0.0126)	(0.0102)	(0.0107)
Number of Reviews prior to observation period (by Restaurant) × Post	0.0004	0.0003		-0.0003	-0.0002	-5.99×10^{-5}	-0.0001
	(0.0003)	(0.0003)		(0.0002)	(0.0002)	(0.0001)	(0.0001)
Fixed-effects							
Restaurant	Yes	Yes		Yes	Yes	Yes	Yes
Month	Yes			Yes		Yes	
Month × Neighborhood		Yes			Yes		Yes
Fit statistics							
Observations	72,213	72,213	40,838	40,838	40,838	40,838	
R^2	0.35	0.38	0.34	0.37	0.35	0.38	
Dependent variable mean	0.71	0.71	0.69	0.69	0.75	0.75	
Dependent variable SD	0.31	0.31		0.27	0.27	0.21	0.21

Notes: This table presents OLS estimates using the same specifications as Table 1, but controlling for the interaction between the total number of reviews pre-treatment and the *post* dummy. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.6: Controlling for Newly Pedestrianized Streets in Paris During the Summer of 2020: The Effect of Tourism on Restaurant-Level Ratings by Parisians

Dependent Variable:	Avg. Rating by Parisians									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Share of Non-Parisians	0.0821***	0.0858***	0.0819***	0.0863***	0.0819***	0.0859***	0.0819***	0.0861***	0.0820***	0.0868***
× Post-Lockdown	(0.0201)	(0.0241)	(0.0201)	(0.0241)	(0.0201)	(0.0241)	(0.0201)	(0.0241)	(0.0201)	(0.0241)
Pedestrian Street (< 50m)			-0.0121	-0.0105						
× Post-Lockdown			(0.0147)	(0.0169)						
Pedestrian Street (< 100m)					-0.0059	-0.0046				
× Post-Lockdown					(0.0114)	(0.0144)				
Pedestrian Street (< 150m)							-0.0086	-0.0055		
× Post-Lockdown							(0.0100)	(0.0134)		
Pedestrian Street (< 200m)									-0.0106	-0.0148
× Post-Lockdown									(0.0092)	(0.0129)
Fixed-effects										
Restaurant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes		Yes		Yes		Yes		Yes	
Month x Quartier		Yes		Yes		Yes		Yes		Yes
Fit statistics										
Observations	72,471	72,471	72,471	72,471	72,471	72,471	72,471	72,471	72,471	72,471
R^2	0.36	0.38	0.36	0.38	0.36	0.38	0.36	0.38	0.36	0.38
Dependent variable mean	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71

Notes: This table presents OLS estimates using the same specifications as Table 1, but controlling for a dummy variable for restaurants located in various buffer zones around newly pedestrianized streets, interacted with the *post* dummy. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.7: Controlling for Distance to Places of Attack

Dependent Variable:	Avg. Rating by Parisians			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews	0.0222**	0.0235**	0.0290**	0.0276**
× Post-Lockdown	(0.0105)	(0.0104)	(0.0120)	(0.0123)
× (log) Distance to				
Bataclan		0.0225**		0.0296
		(0.0111)		(0.0186)
Stade de France		0.0210		0.0200
		(0.0150)		(0.0607)
Petit Cambodge		-0.0162		-0.0408*
		(0.0182)		(0.0238)
Bonne Bière		-0.0067		-0.0176
		(0.0222)		(0.0152)
La Belle Equipe		-0.0146		-0.0216*
		(0.0181)		(0.0118)
Comptoir Voltaire		0.0092		0.0321
		(0.0161)		(0.0230)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes	Yes		
Month x Neighborhood			Yes	Yes
Fit statistics				
Observations	42,033	42,033	42,033	42,033
R^2	0.36	0.36	0.39	0.39
Dependent variable mean	0.69	0.69	0.69	0.69
Dependent variable SD	0.27	0.27	0.27	0.27

Notes: This table presents OLS estimates using the same specifications as Table 1, but includes controls for distances to the sites of terrorist attacks. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.8: Accounting for Skewness

Natural experiments:	**Before and After First Pandemic Lockdown** *(Post = Post-Lockdown)*			**Before and After Terrorist Attack – November 2015** *(Post = Post-Terrorist Attack)*			
Dependent variables:	Avg. Rating by Parisians			Avg. Rating by Parisians		Avg. Rating by Non-Parisians	
	(1)	(2)		(3)	(4)	(5)	(6)
Panel A: Excluding top 1% Users by Reviews							
Share of Non-French Reviews prior to observation period (by Restaurant) × Post	0.0862***	0.1092***		0.0249*	0.0400**	0.0010	-0.0090
	(0.0280)	(0.0374)		(0.0130)	(0.0157)	(0.0111)	(0.0126)
Observations	54,424	54,424	30,581	30,581	30,581	30,581	
R^2	0.40	0.43	0.38	0.42	0.40	0.44	
Panel B: Excluding top 1% Restaurants by Reviews							
Share of Non-French Reviews prior to observation period (by Restaurant) × Post	0.0737***	0.0801***		0.0236*	0.0300**	-0.0056	-0.0110
	(0.0226)	(0.0274)		(0.0119)	(0.0132)	(0.0106)	(0.0114)
Observations	72,762	72,762	38,937	38,937	38,937	38,937	
R^2	0.36	0.38	0.36	0.39	0.36	0.40	
Fixed-effects							
Restaurant	Yes	Yes		Yes	Yes	Yes	Yes
Month	Yes			Yes		Yes	
Month × Neighborhood		Yes			Yes		Yes

Notes: This table presents OLS estimates using the same specifications as Table 1, but applied to restricted samples. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.9: Neighborhood Resident Characteristics Heterogeneity

Dependent Variable:	Rating by Parisians				
	(1)	(2)	(3)	(4)	(5)
Share of Non-French Reviews × Post-Lockdown	0.0858***	0.0852***	0.0887***	0.0800***	0.0839***
	(0.0255)	(0.0256)	(0.0255)	(0.0283)	(0.0309)
Share of Non-French Reviews × Post-Lockdown × Older Age Group Share		0.0065			
		(0.0221)			
Share of Non-French Reviews × Post-Lockdown × Younger Age Group Share			-0.0241		
			(0.0231)		
Share of Non-French Reviews × Post-Lockdown × Poorer Group Share					0.0142
					(0.0405)
Fixed-effects					
User	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	72,471	72,471	72,471	63,702	63,702
R^2	0.38	0.38	0.38	0.38	0.38
Dependent variable mean	0.71	0.71	0.71	0.72	0.72
Dependent variable SD	0.31	0.31	0.31	0.31	0.31

Notes: This table presents OLS estimates using the same specifications as Table 2, with additional interactions with neighborhood Census-based characteristics. The unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-Lockdown* variable is a dummy variable, activated after the first pandemic lockdown. Standard errors are clustered at the neighborhood level. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.10: Price Heterogeneity

Dependent Variable:	Avg. Rating by Parisians			
	(1)	(2)	(3)	(4)
Share of Non-French Reviews	0.0837***	0.0891***		
× Post-Lockdown	(0.0212)	(0.0263)		
× Low Price (N=2091)			0.0182	0.0232
			(0.0603)	(0.0647)
× Medium Price (N=7518)			0.1029***	0.1131***
			(0.0215)	(0.0286)
× High Price (N=643)			-0.0022	-0.0108
			(0.0493)	(0.0466)
Post-Lockdown × Low Price			0.0154	0.0067
			(0.0176)	(0.0192)
Post-Lockdown × High Price			0.0378*	0.0432*
			(0.0192)	(0.0222)
Fixed-effects				
Restaurant	Yes	Yes	Yes	Yes
Month	Yes		Yes	
Month × Neighborhood		Yes		Yes
Fit statistics				
Observations	71,630	71,630	71,630	71,630
R^2	0.35	0.38	0.35	0.38
Dependent variable mean	0.72	0.72	0.72	0.72
Dependent variable SD	0.31	0.31	0.31	0.31

Notes: This table presents OLS estimates using the same specifications as Table 1, with additional interactions with restaurant price categories. In all columns, the unit of analysis is a month × restaurant pair. The table displays results for two notable shocks in tourism: the first year of the pandemic and the 2015 terrorist attack. The dependent variable in Columns 1-4 is the average rating of restaurants among users with a home location in Paris. For the terrorist attack, we additionally measure the effect on ratings by tourists in Columns 5-6. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019 and up to 2014, corresponding to the pandemic and terrorism-induced shocks, respectively. The *Post* variable is a dummy, activated after each shock. Standard errors clustered at the neighborhood level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.11: Tourism and Ratings: Different Clustering

Dependent variables:	Avg. Rating by Parisians		
	(1)	(2)	(3)
Share of Non-French Reviews × Post-Lockdown	0.0858***	0.0858***	0.0858***
	(0.0255)	(0.0249)	(0.0241)
Fixed-effects			
Restaurant	Yes	Yes	Yes
Month x Neighborhood	Yes	Yes	Yes
Clustering			
	Neighborhood	Grid cell	Restaurant
Fit statistics			
Observations	72,471	72,471	72,471
R^2	0.38	0.38	0.38
Dependent variable mean	0.71	0.71	0.71

Notes: This table presents OLS estimates. In all columns, the unit of analysis is a month × restaurant pair. The dependent is the average rating of restaurants among users with a home location in Paris. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-Lockdown* variable is a dummy, activated after the first pandemic lockdown. Standard errors clustered at a specified level are in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

D Text Analysis

Table D.1: Dictionary 1: Words, Phrases and Wildcard Expressions for Dictionary-Based Labelling of Reviews

Low Food Quality
pas bon *(not tasty)*, sans goût *(no taste)*, aucun saveur *(no taste)*, réchauff *(reheated)*, pas très bon *(not very tasty)*, aucun goût *(no taste)*, cuisine bof *(kitchen yuck)*, mauvaise cuisson *(poor cooking)*, goût bizzare *(wierd taste)*, industriel *(industrial)*, avarié *(rotten)*, pas assez cuit *(undercooked)*, trop cuit *(overcooked)*, supermarch *(supermarket)*, tombé malade *(got sick)*, pas cuit *(not cooked)*, sans saveur *(no taste)*, vomir *(vomit)*, mauvaise qualité *(bad quality)*, indigestion *(indigestion)*, intoxication *(intoxication)*, pas frais *(not fresh)*, fade *(no taste)*, surgel *(frozen food)*, insipid *(no taste)*, dégueulass *(disgusting)*, micro-ond *(microwaved)*, pas fait maison *(not "homemade")*
Too Expensive
prix élevés *(high price)*, cher *(expensive)*, prix sont élevés *(high price)*, prix sont très élevés *(very high price)*
Too Noisy
bruyant *(noisy)*, beaucoup de bruit *(a lot of noise)*
Long Wait
long *(long)*, lent *(long)*
Tourism
touris *(tourist/tourism)*

Table D.2: Dictionary 2: Seeding Words for Word Embedding Labelling of Reviews

Low Food Quality
insipide *(no taste)*, indigestion *(indigestion)*, surgelé *(frozen food)*, industriel *(industrial)*, fade *(no taste)*
Too Expensive
cher *(expensive)*, élevé *(increased)*, excessif *(excessive)*, coûteux *(expensive)*, onéreux *(expensive)*
Too Noisy
bruyant *(noisy)*, bruit *(noise)*
Long Wait
long *(long)*, lent *(long)*, tardif *(late)*
Tourism
touriste *(tourist)*, tourisme *(tourism)*

Table D.3: Ratings and Textual Variables (Dictionary-Based)

Dependent Variable:	Rating by Parisian				
	(1)	(2)	(3)	(4)	(5)
$\mathbb{1}$\{tourists\}	-0.0850***				
	(0.0092)				
$\mathbb{1}$\{poor food\}		-0.2905***			
		(0.0050)			
$\mathbb{1}$\{expensive\}			-0.1109***		
			(0.0058)		
$\mathbb{1}$\{noisy\}				-0.0540***	
				(0.0070)	
$\mathbb{1}$\{long wait\}					-0.1071***
					(0.0072)
Fixed-effects					
User	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes
Date	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	112,238	112,238	112,238	112,238	112,238
R^2	0.75	0.77	0.75	0.74	0.75
Dependent variable mean	0.72	0.72	0.72	0.72	0.72

Notes: This table presents OLS estimates. In all columns, the unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The explanatory variables are dummies that indicate whether the text of a review relates to the given topic (dictionary-based). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table D.4: Ratings and Textual Variables (Embedding-Based)

Dependent Variable:	Rating by Parisian				
	(1)	(2)	(3)	(4)	(5)
Tourism (cosine dist.)	-1.011***				
	(0.0222)				
Poor food (cosine dist.)		-1.276***			
		(0.0127)			
Expensive (cosine dist.)			-0.5729***		
			(0.0204)		
Noisy (cosine dist.)				-0.8899***	
				(0.0186)	
Long wait (cosine dist.)					-1.252***
					(0.0242)
Fixed-effects					
User	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes
Date	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	112,238	112,238	112,238	112,238	112,238
R^2	0.76	0.81	0.75	0.76	0.77
Dependent variable mean	0.72	0.72	0.72	0.72	0.72

Notes: This table presents OLS estimates. In all columns, the unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The explanatory variables are cosine distances to the centroids corresponding to the given topic (word embedding). Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table D.5: Effect on Ratings, Controlling for Text Variables (Embedding-Based)

Dependent Variable:	Rating by Parisian					
	(1)	(2)	(3)	(4)	(5)	(6)
Share of Non-French Reviews	0.0929**	0.0555	0.0674**	0.0698*	0.0684*	0.0694*
× Post-Lockdown	(0.0413)	(0.0394)	(0.0332)	(0.0403)	(0.0400)	(0.0385)
Tourism (cosine dist.)		-0.9731***				
		(0.0234)				
Poor food (cosine dist.)			-1.260***			
			(0.0136)			
Expensive (cosine dist.)				-0.5786***		
				(0.0214)		
Noisy (cosine dist.)					-0.8668***	
					(0.0200)	
Long wait (cosine dist.)						-1.232***
						(0.0249)
Fixed-effects						
User × Post-Lockdown	Yes	Yes	Yes	Yes	Yes	Yes
Restaurant	Yes	Yes	Yes	Yes	Yes	Yes
Month × Neighborhood	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	112,238	112,238	112,238	112,238	112,238	112,238
R^2	0.77	0.78	0.83	0.77	0.78	0.79
Dependent variable mean	0.72	0.72	0.72	0.72	0.72	0.72
Dependent variable SD	0.33	0.33	0.33	0.33	0.33	0.33

Notes: This table presents OLS estimates using the same specifications as Table 2, with additional controls derived from review texts, calculated as cosine distances to pre-defined poles. The unit of analysis is an individual review. The dependent variable is the rating of a review left by a Parisian user. All ratings are scaled from 0 to 1. The share of non-French reviews is calculated for the periods up to 2019. The *Post-Lockdown* variable is a dummy variable, activated after the first pandemic lockdown. Standard errors are clustered at the neighborhood level. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

E Data Availability

Please see the replication package for the code producing tables and figures contained in the paper Avetian and Pauly (2025).